



LECTURE 06 WAVELETS AND MULTIREOLUTION PROCESSING

Punnarai Siricharoen, Ph.D.

OBJECTIVES

- To be able to understand the basic concept of wavelets (Haar wavelet)
- To be able to apply wavelets transform for multiresolution analysis

CONTENT

- Introduction
- Image Pyramid
- Fast Wavelet Transform (1D & 2D)
- Python Exercises

INTRODUCTION

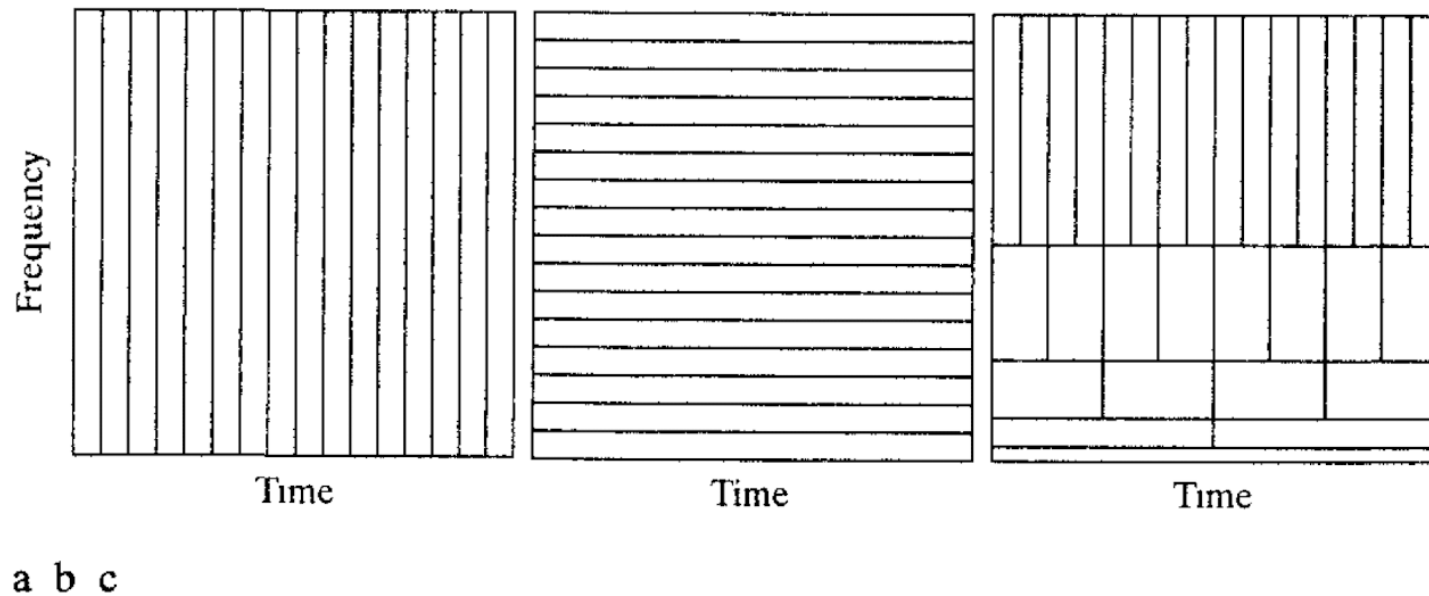
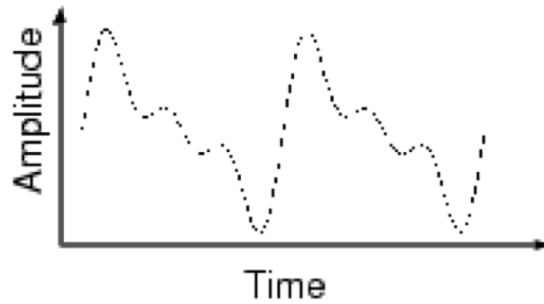
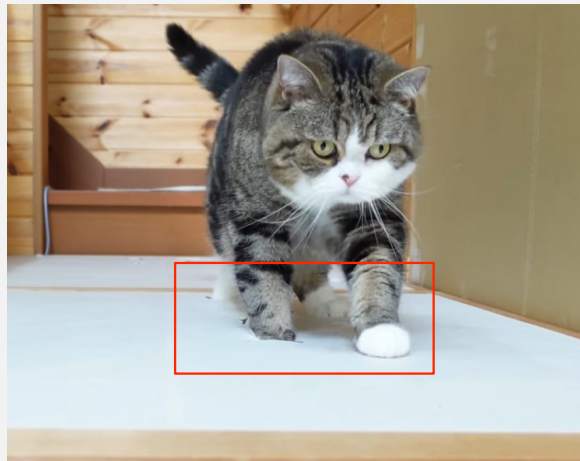


FIGURE 7.21 Time-frequency tilings for (a) sampled data, (b) FFT, and (c) FWT basis functions.

INTRODUCTION

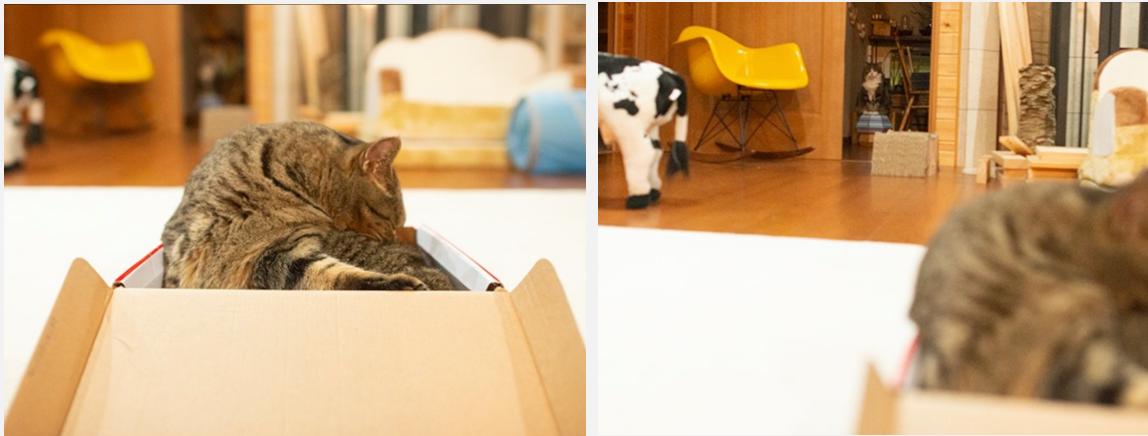
- Small object, low contrast – make it higher resolution
- Large object, high contrast – look at coarse view
- Motivation for multiresolution processing
 - All of them together – study them in several resolutions



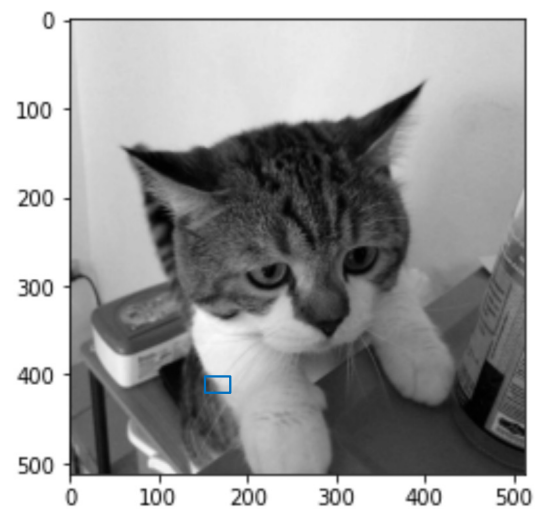
แฟนคลับแมวกล่อง

INTRODUCTION

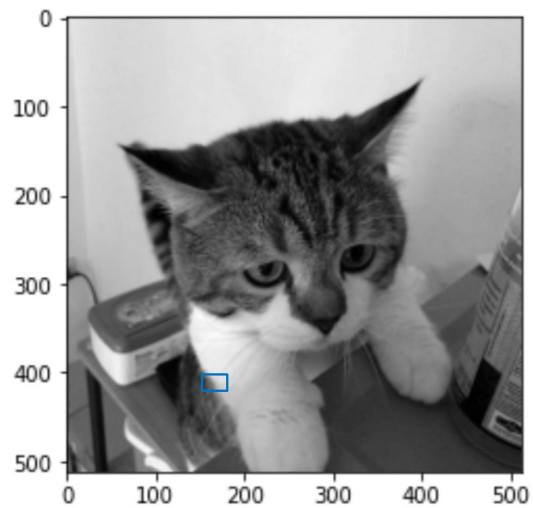
- Small object, low contrast – make it higher resolution
- Large object, high contrast – look at coarse view
- Motivation for multiresolution processing
 - All of them together – study them in several resolutions



แฟนคลับแมวกลอง



```
[ [ 67  66  70  87  98 114 138 147 151 166 180 183 180 181 186 185]
  [ 56  61  75  87  96 110 127 132 138 155 174 179 178 180 184 184]
  [ 57  68  85  88  89  96 108 115 124 140 156 168 174 177 178 176]
  [ 55  65  77  77  77  84  98 111 122 132 149 163 171 176 179 179]
  [ 48  53  63  63  70  83  97 106 114 122 136 149 157 165 172 175]
  [ 43  50  57  57  65  80  89  90  98 112 118 134 149 162 171 172]
```



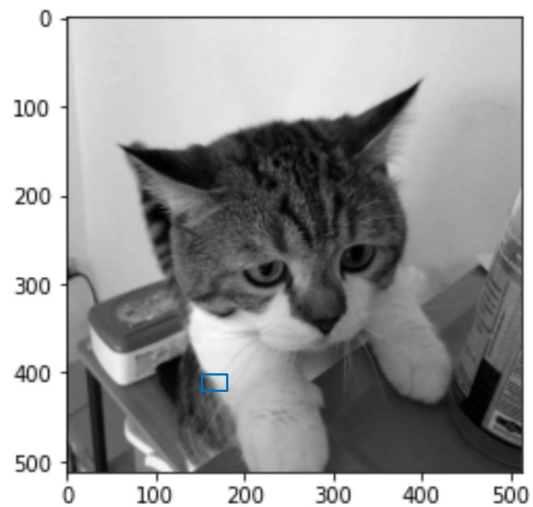
[	67	66	70	87	98	114	138	147	151	166	180	183	180	181	186	185]
[	56	61	75	87	96	110	127	132	138	155	174	179	178	180	184	184]
[	57	68	85	88	89	96	108	115	124	140	156	168	174	177	178	176]
[	55	65	77	77	77	84	98	111	122	132	149	163	171	176	179	179]
[	48	53	63	63	70	83	97	106	114	122	136	149	157	165	172	175]
[	43	50	57	57	65	80	89	90	98	112	118	134	149	162	171	172]

average

88.5

difference

-1



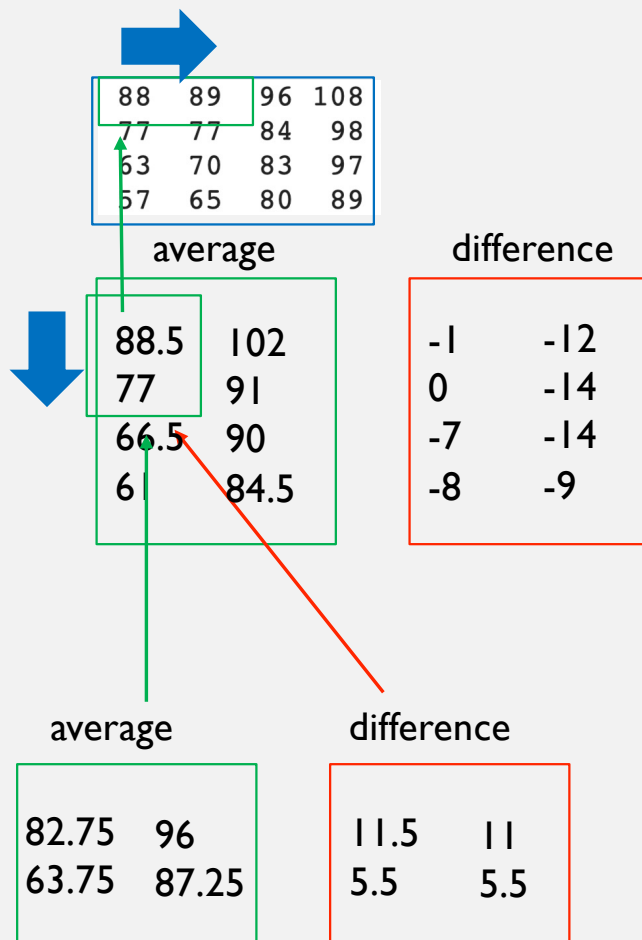
```
[ [ 67  66  70  87  98 114 138 147 151 166 180 183 180 181 186 185]
  [ 56  61  75  87  96 110 127 132 138 155 174 179 178 180 184 184]
  [ 57  68  85  88  89  96 108 115 124 140 156 168 174 177 178 176]
  [ 55  65  77  77  77  84  98 111 122 132 149 163 171 176 179 179]
  [ 48  53  63  63  70  83  97 106 114 122 136 149 157 165 172 175]
  [ 43  50  57  57  65  80  89  90  98 112 118 134 149 162 171 172]
```

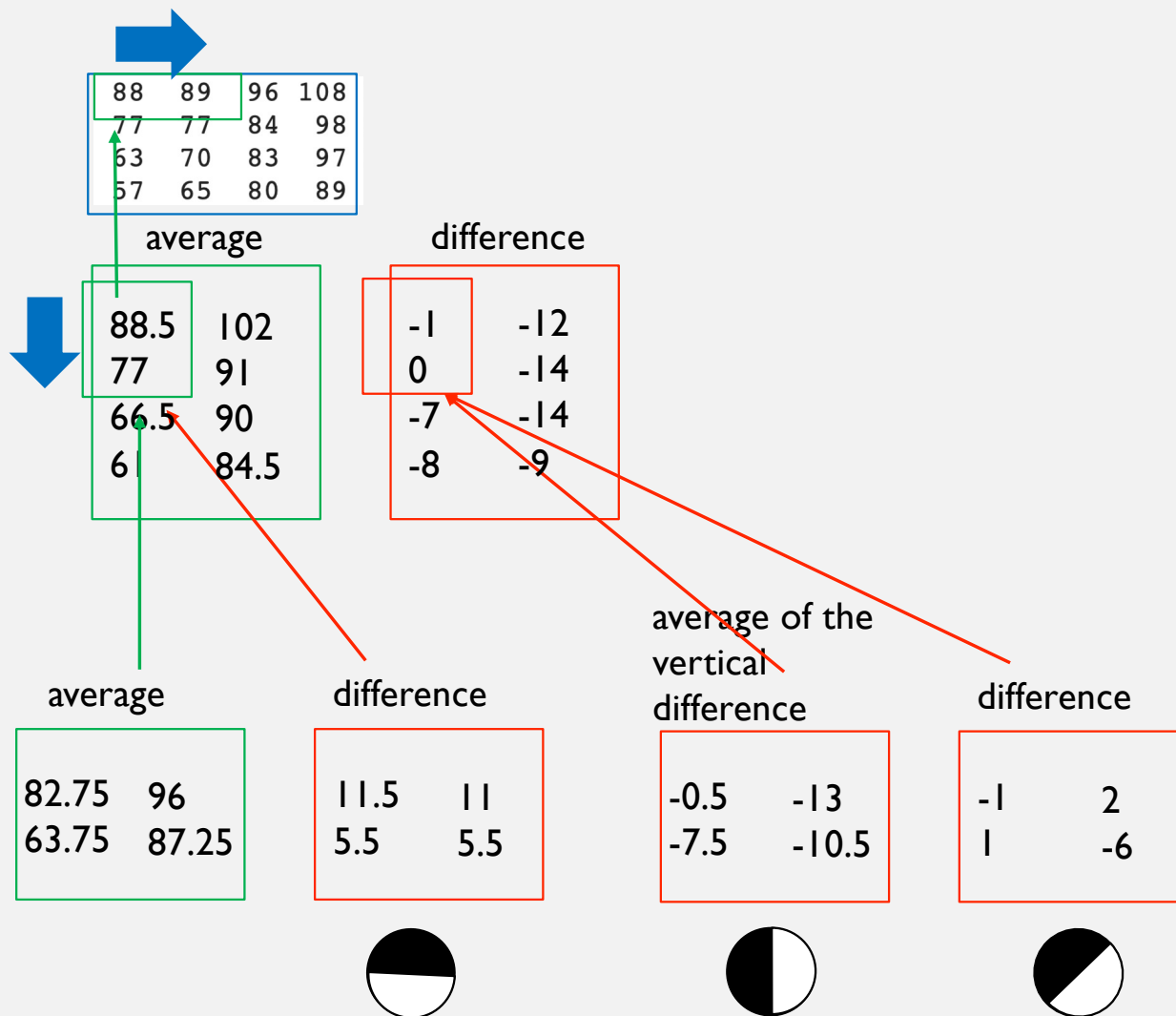
average

88.5	102
77	91
66.5	90
61	84.5

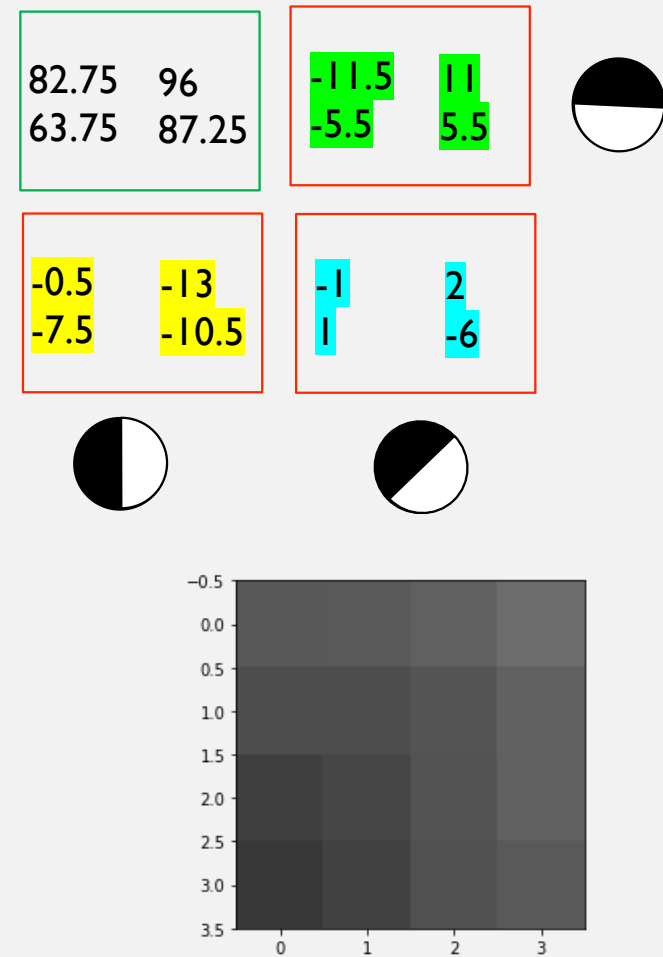
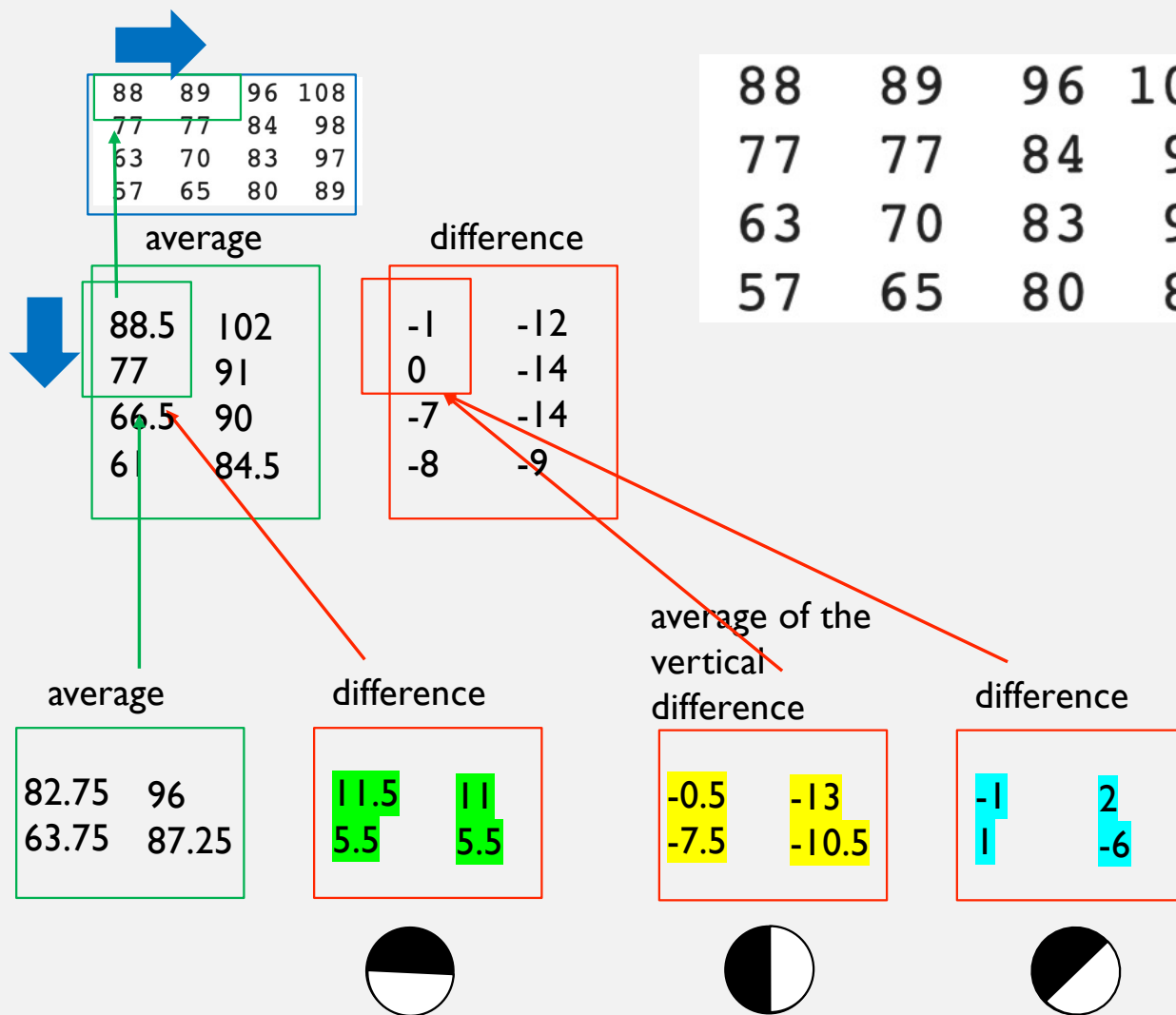
difference

-1	-12
0	-14
-7	-14
-8	-9

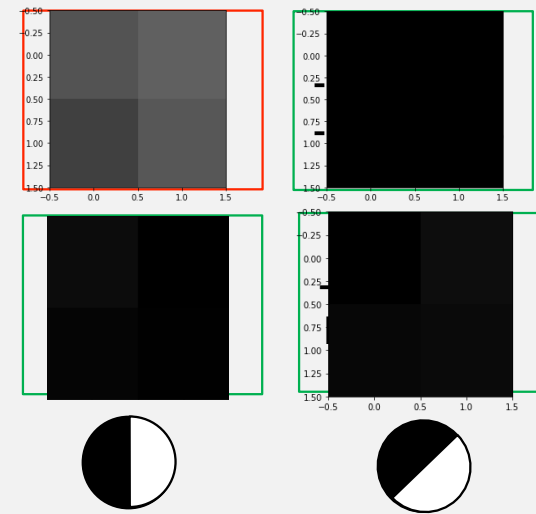
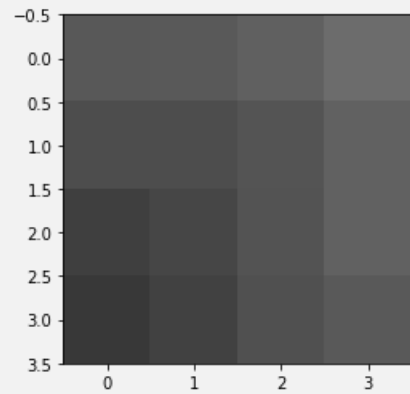
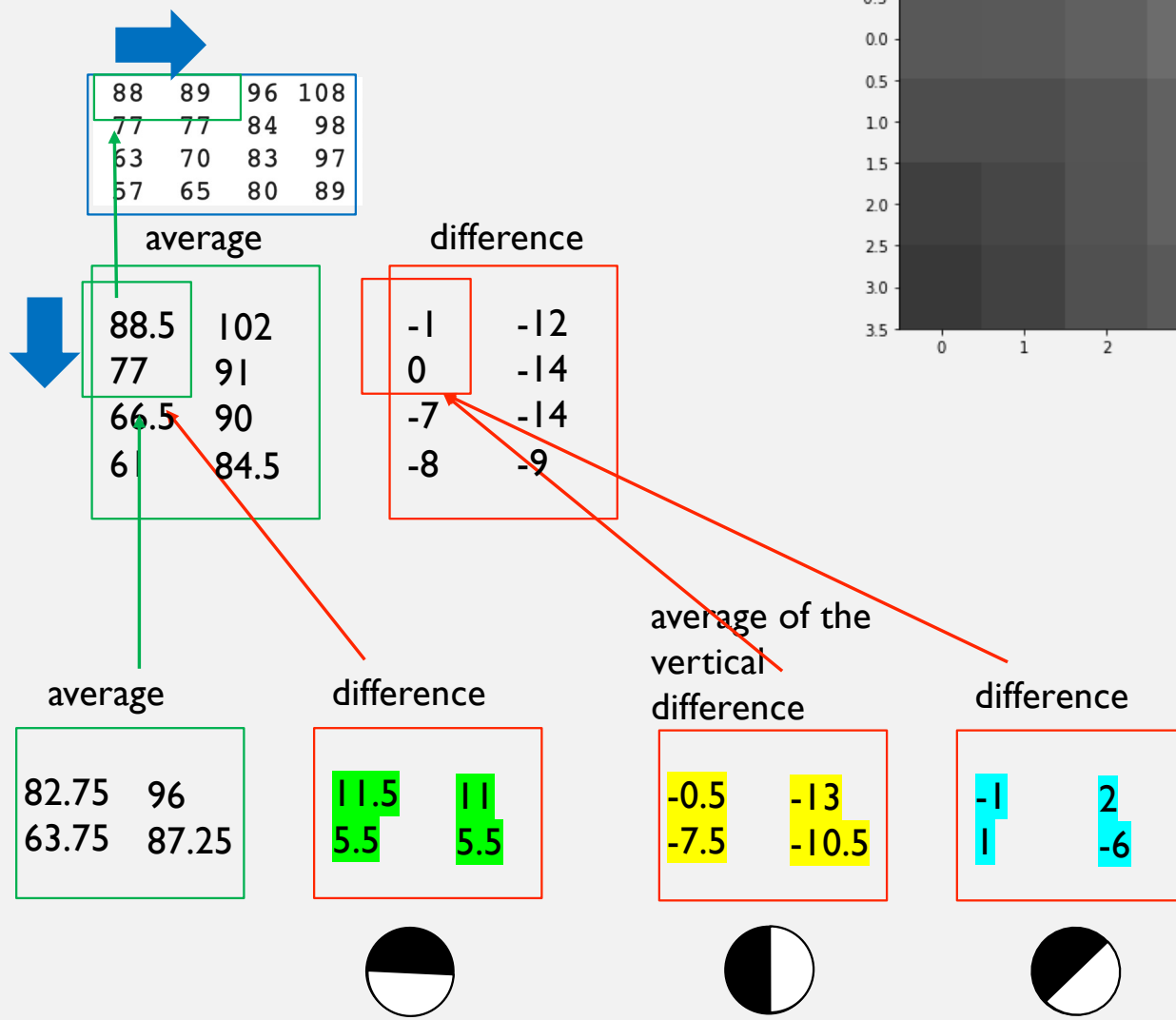




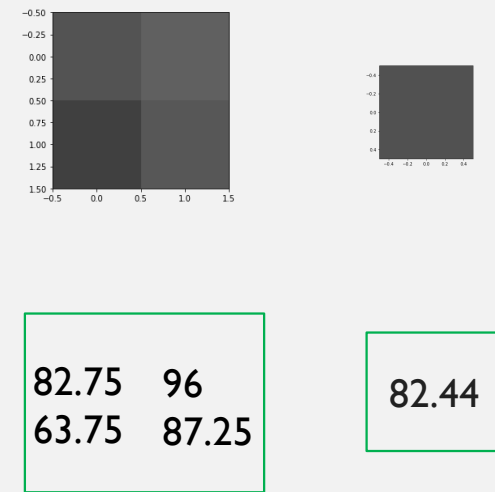
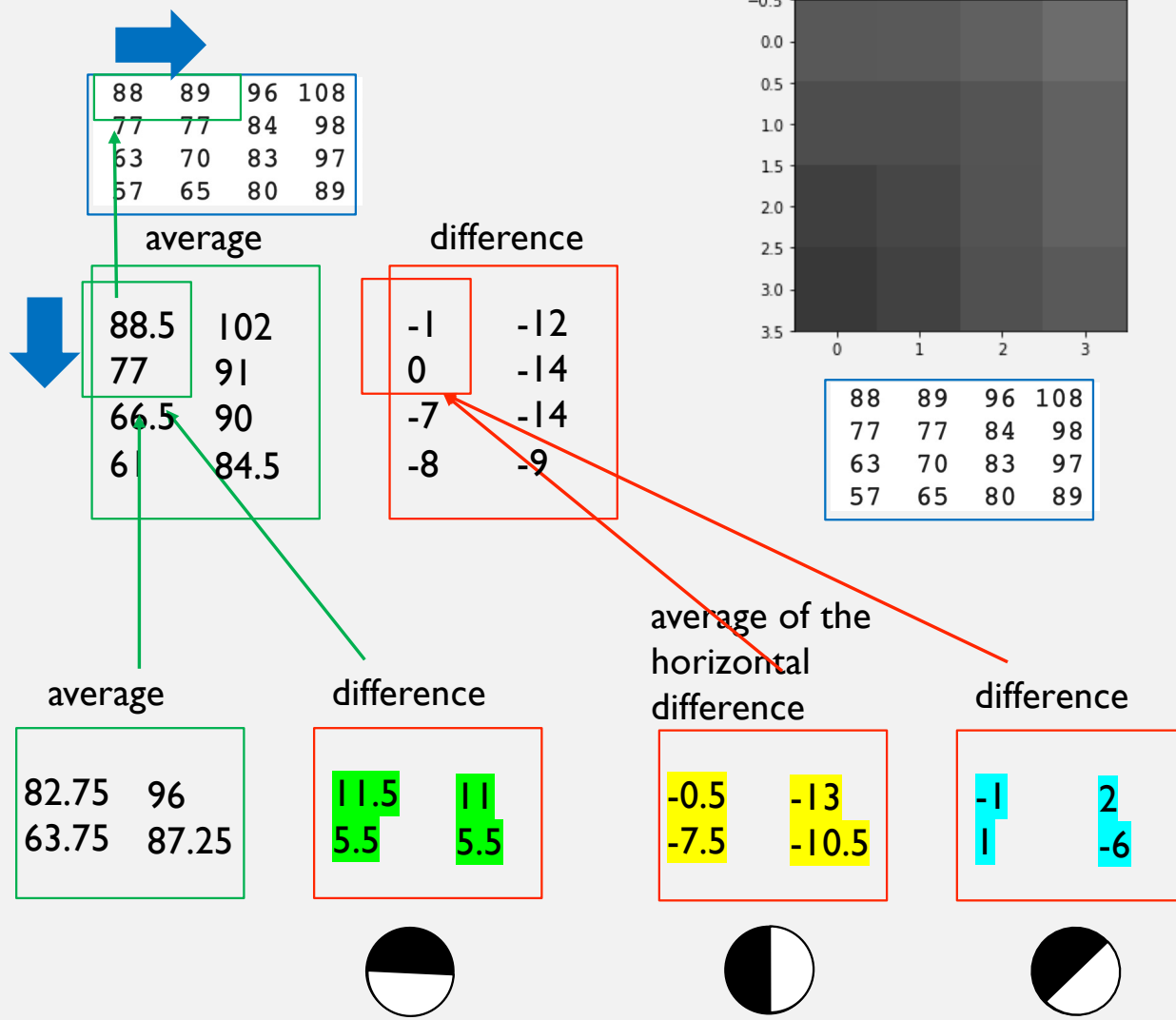
Three species of differences



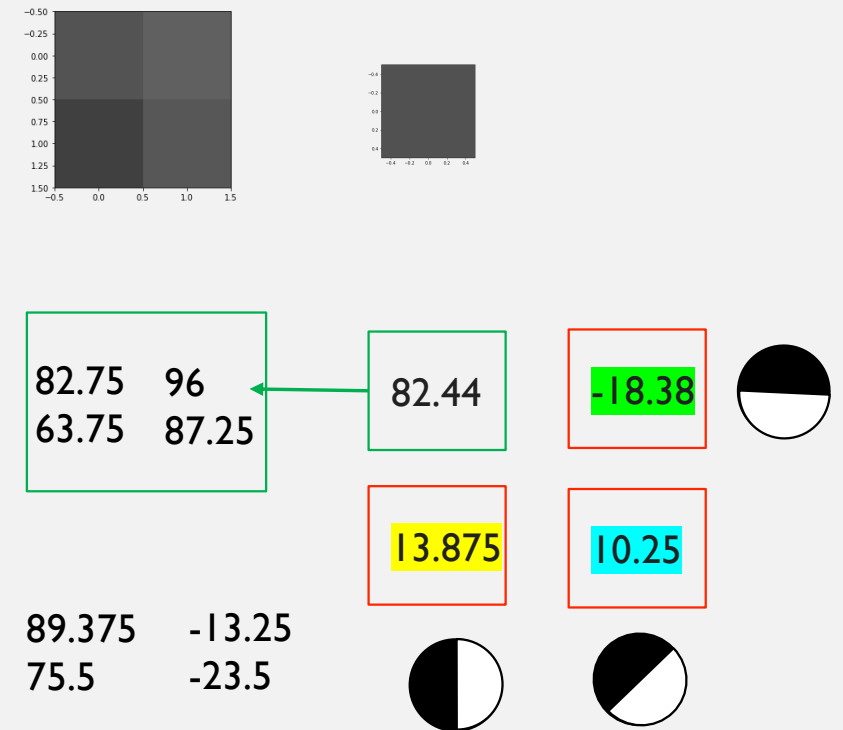
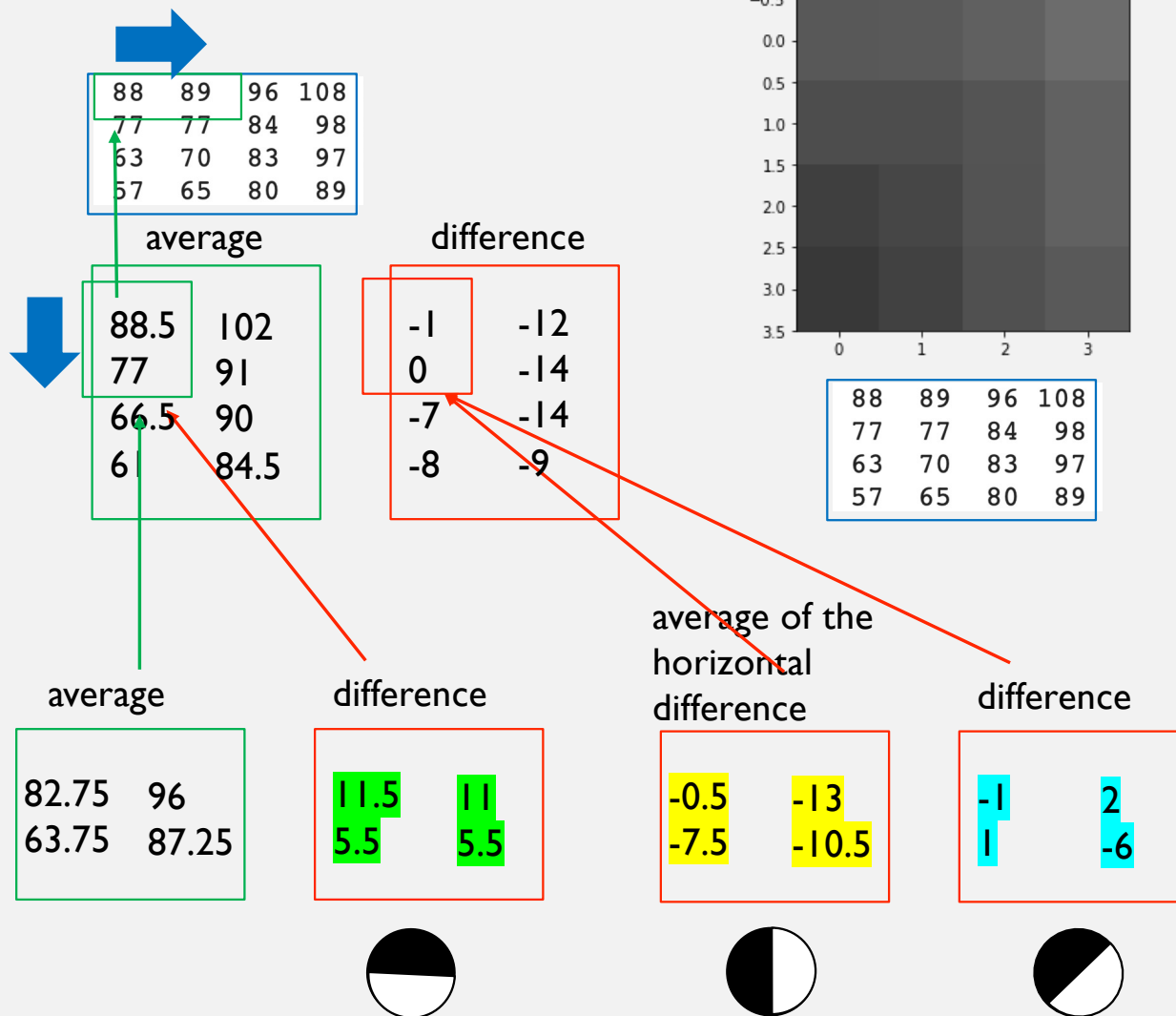
Three species of differences



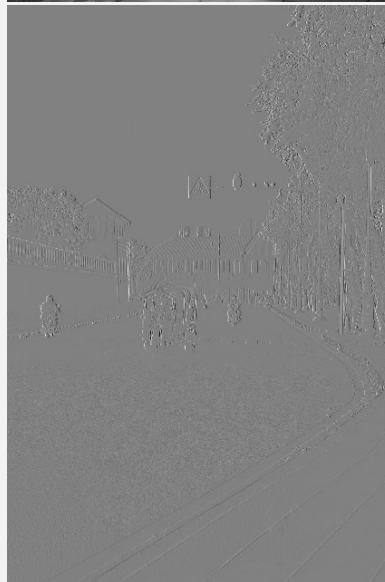
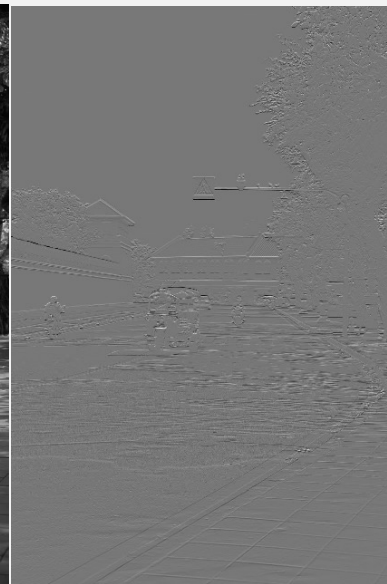
Three species of differences

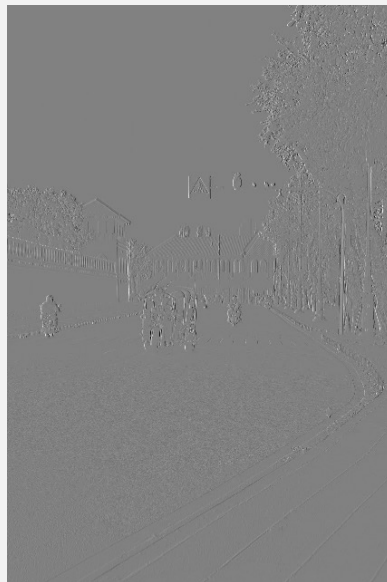
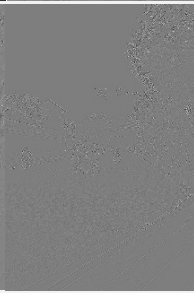
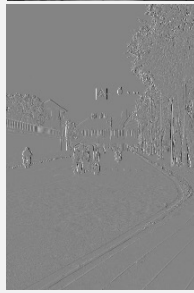
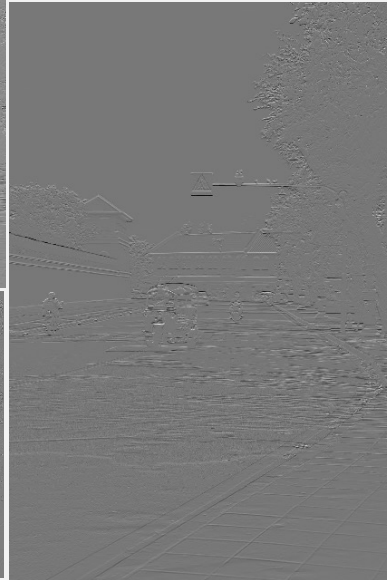
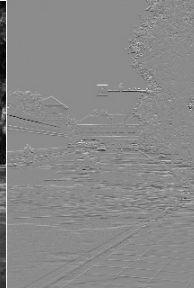


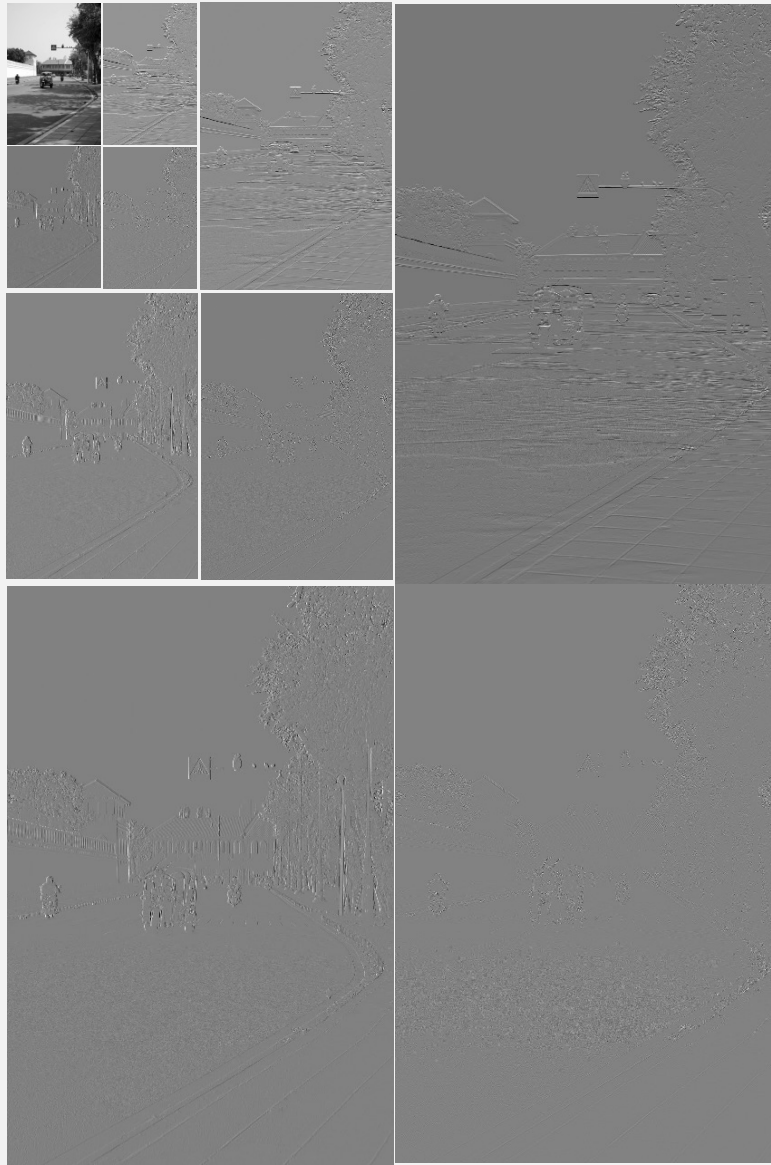
Three species of differences

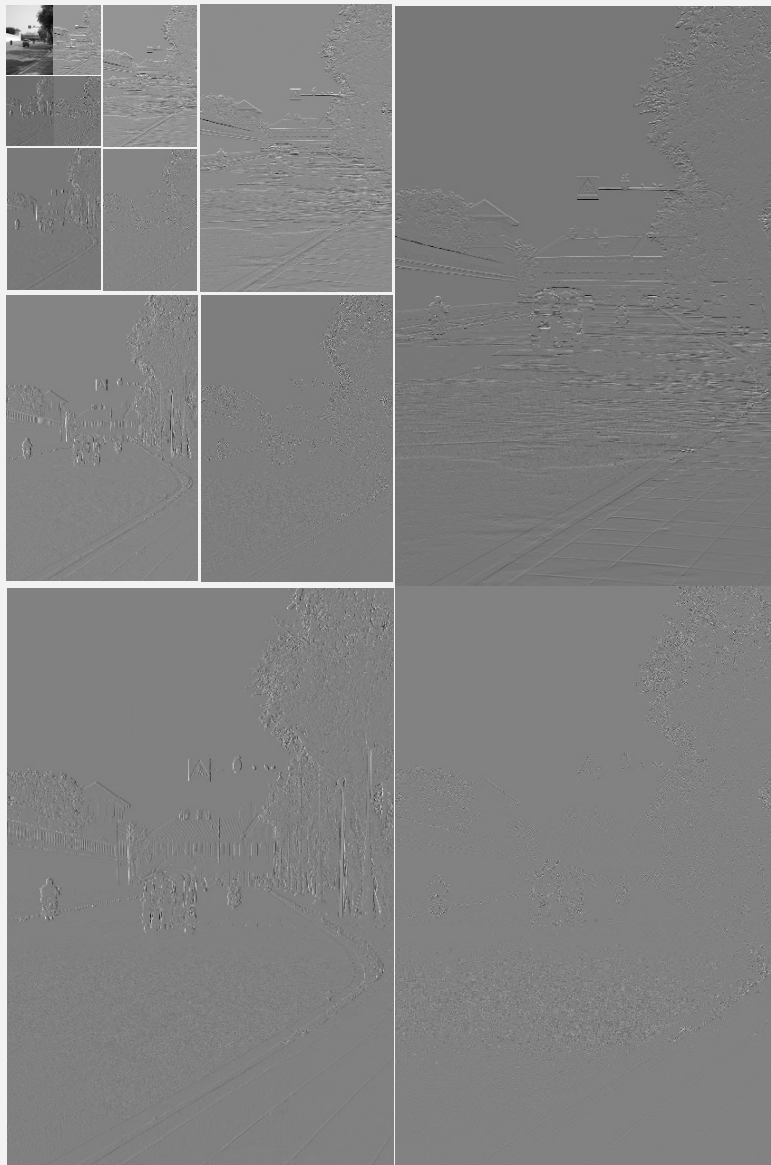


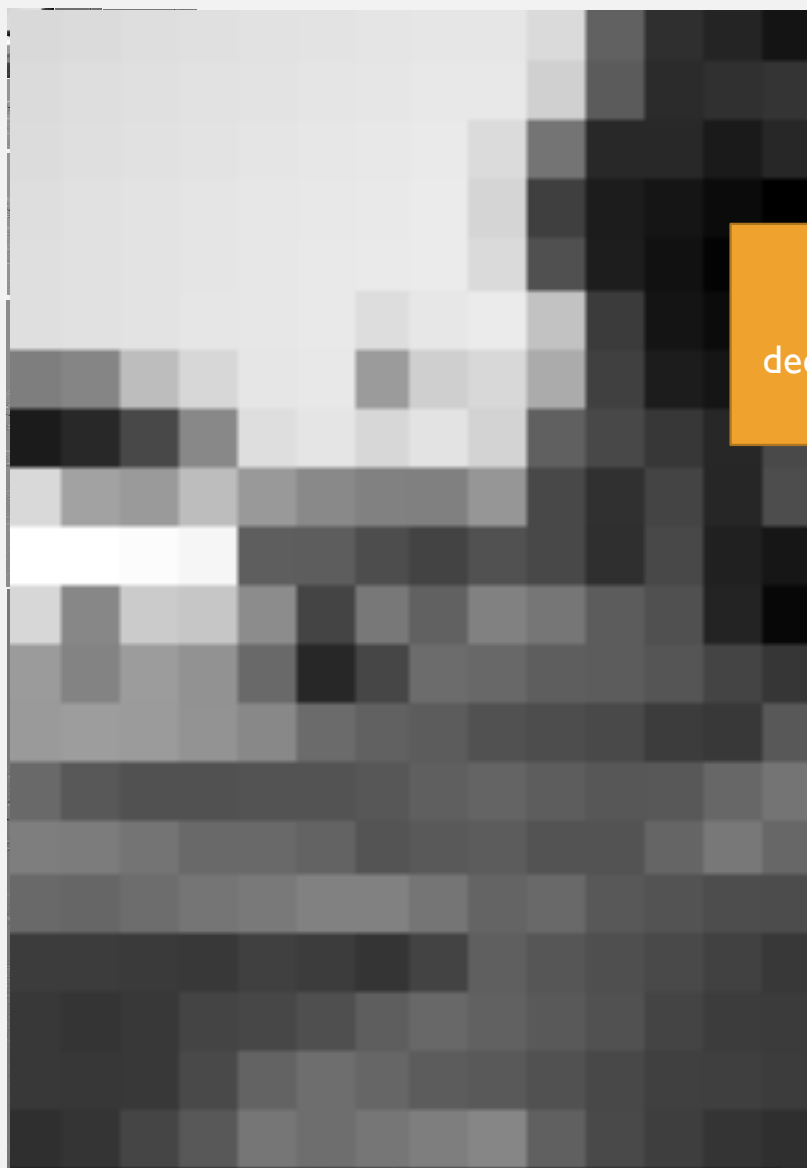
Three species of differences



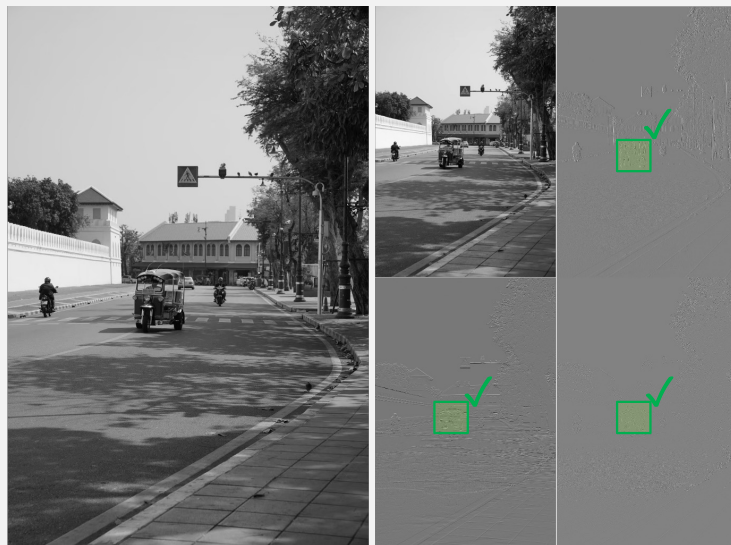








Six-scale
decomposition



Original image

All details are removed except for Tuk-Tuk



All details of Tuk-Tuk are removed

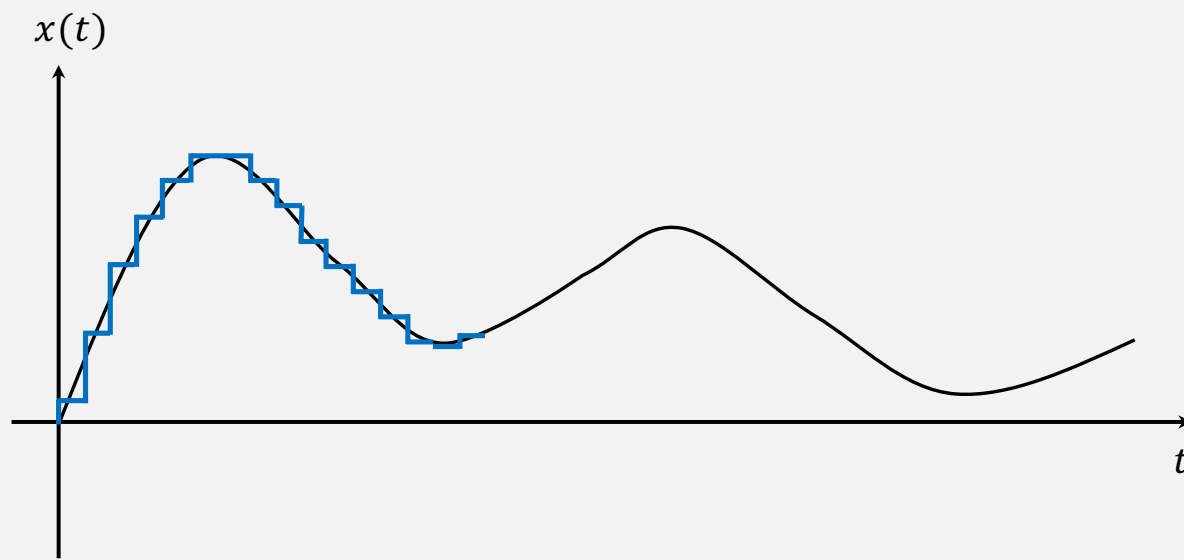


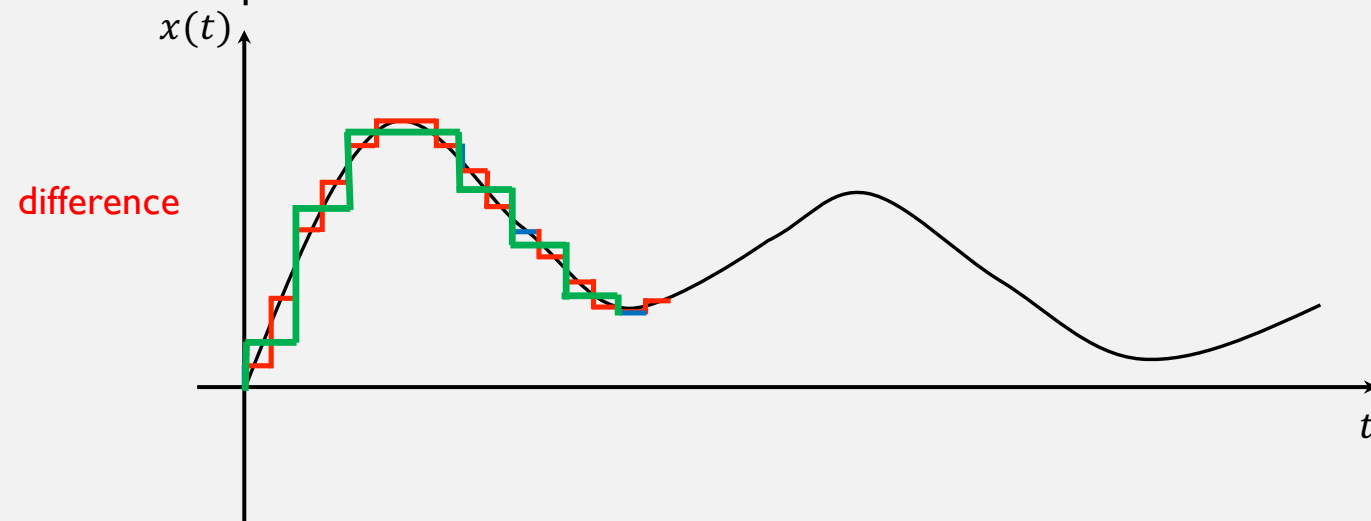
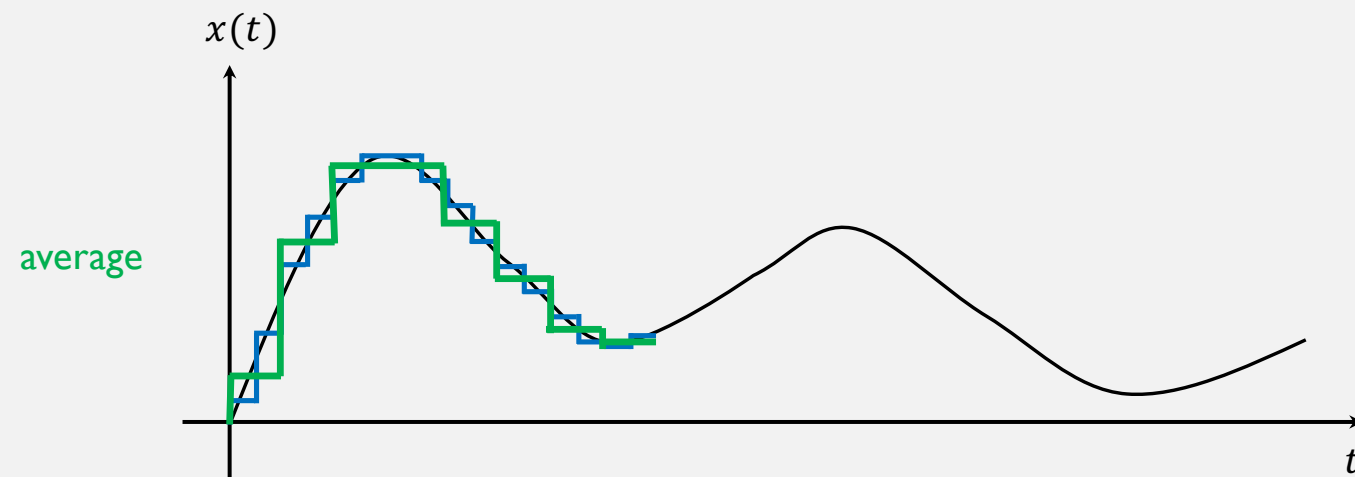


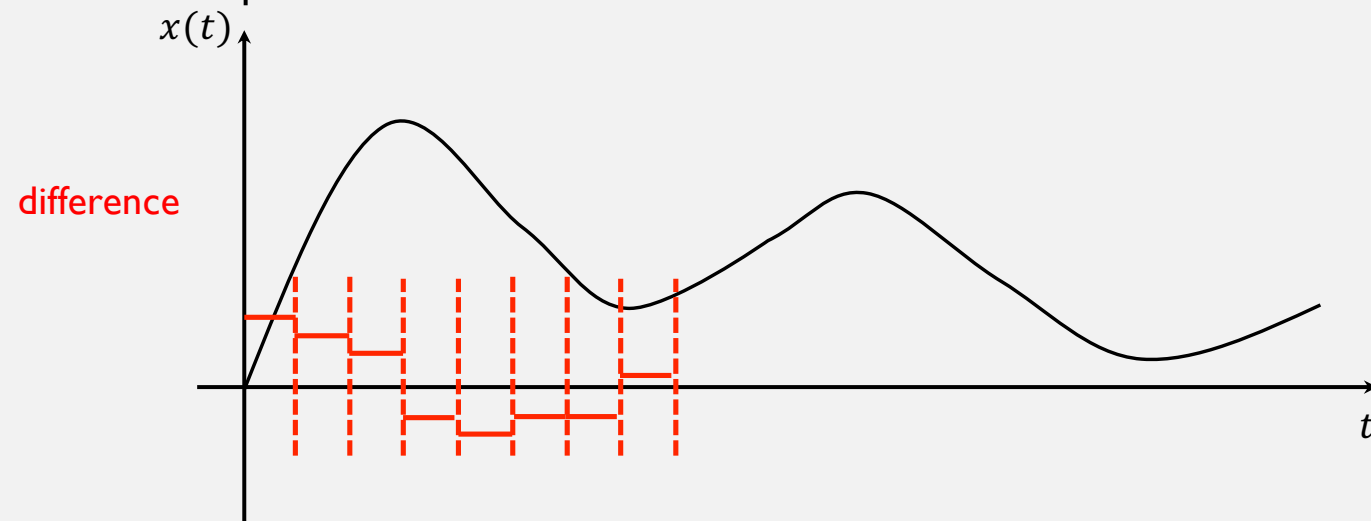
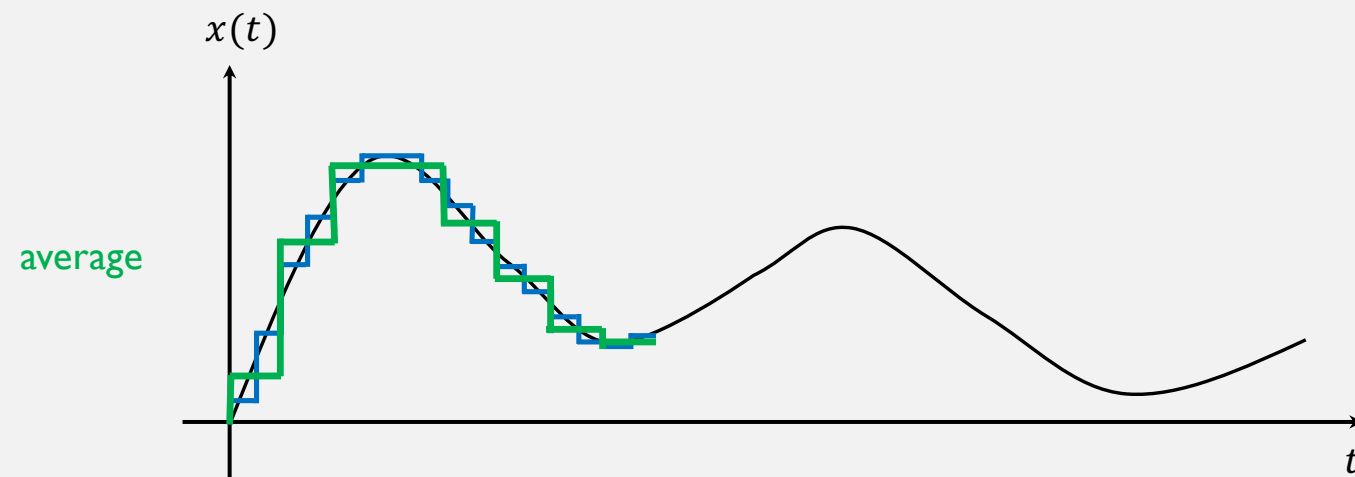
Remove "ES"
details



Remove edge
of the glass
details







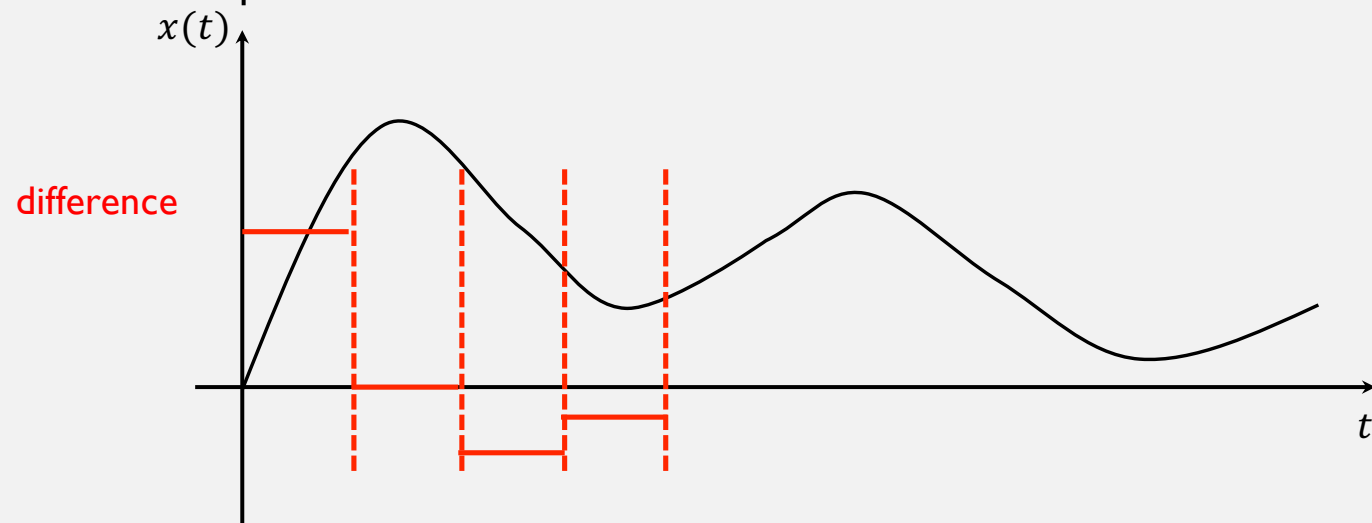
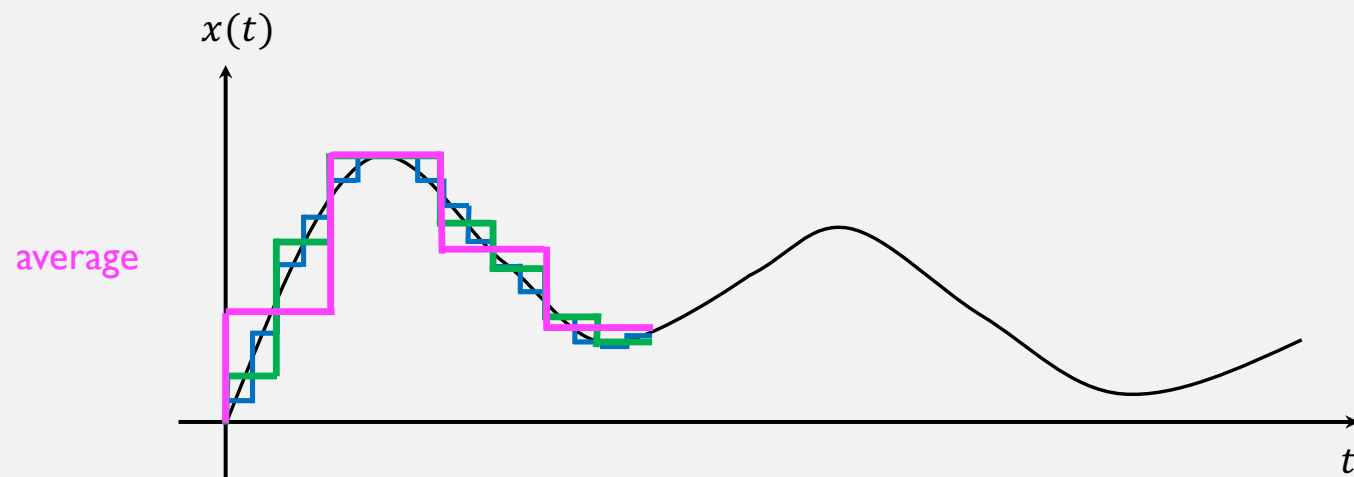


IMAGE PYRAMIDS

- Representing an image more than one resolution
 - Base – high resolution
 - Apex – low resolution
 - Base level J is size $2^J \times 2^J$ (or $N \times N$) ($J = \log_2 N$)
 - Level j is size $2^j \times 2^j$, where $0 \leq j \leq J$
 - Fully populated pyramid – from $2^J \times 2^J$ to $2^0 \times 2^0$ (1×1 too small)
 - Most pyramid truncated to $P+1$ levels ($j = J-P, J-P+1, \dots, J$)

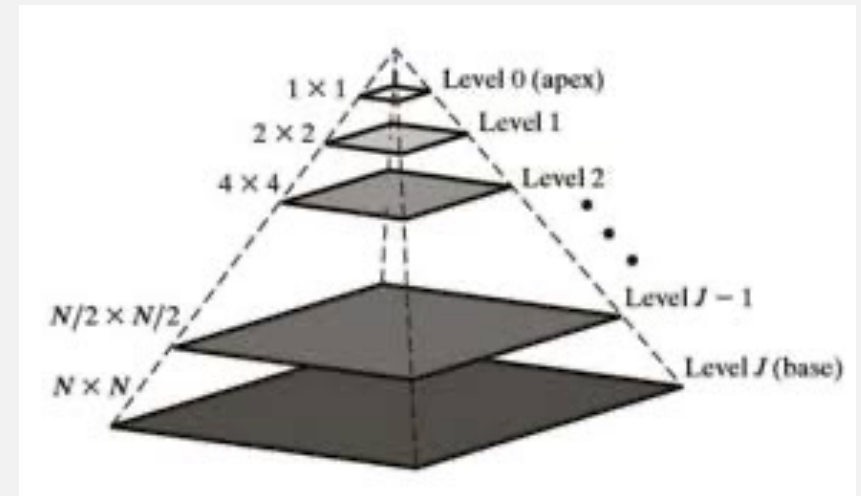


Image from Gonzalez & Woods, Digital Image Processing

IMAGE PYRAMIDS

- System block diagram to create a pyramid

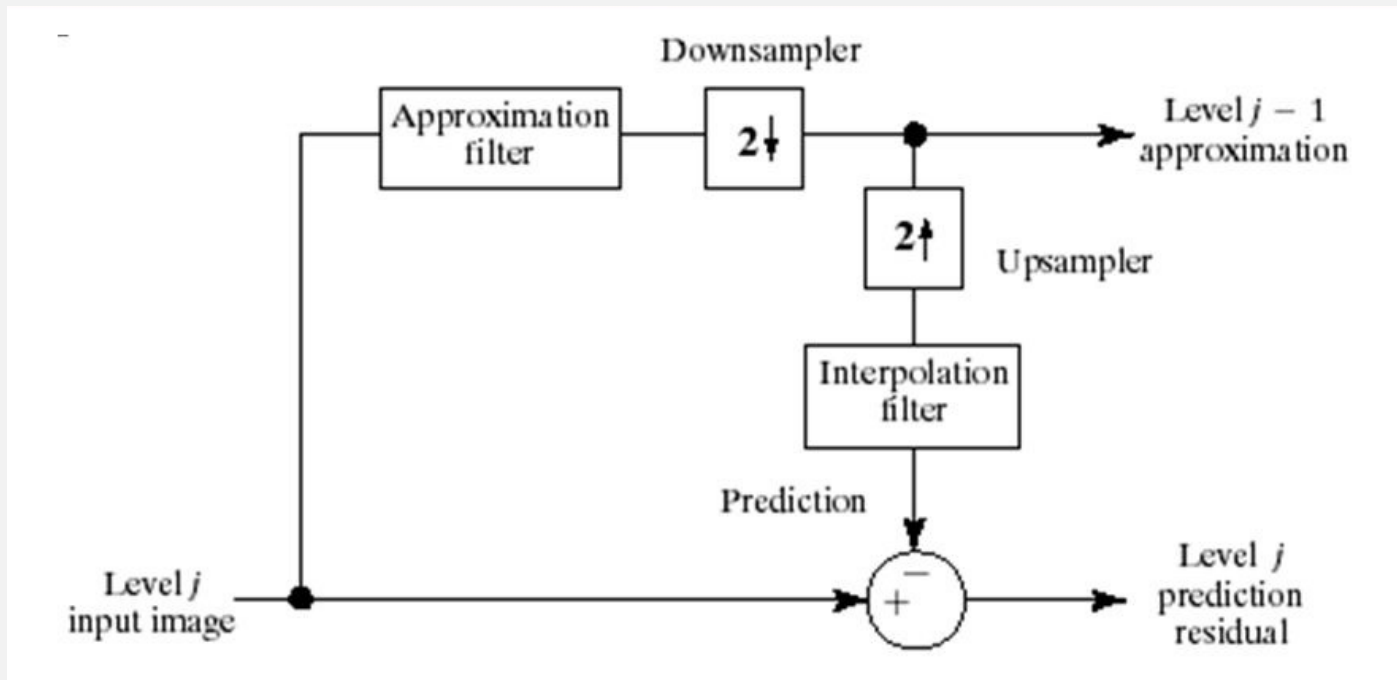


Image from Gonzalez & Woods, Digital Image Processing

Filtering

- averaging filtering creates a mean pyramid
 - lowpass Gaussian filtering creates a Gaussian pyramid
- Without filtering, aliasing can be occurred in the upper levels. (subsampling pyramid)

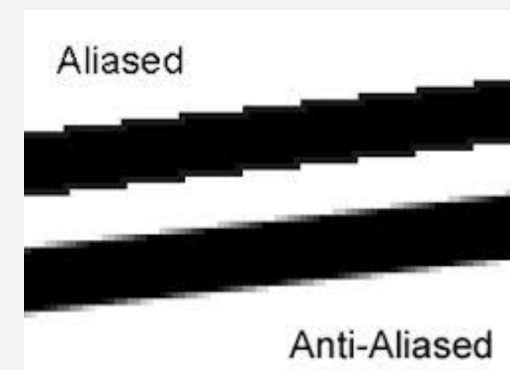


IMAGE PYRAMIDS

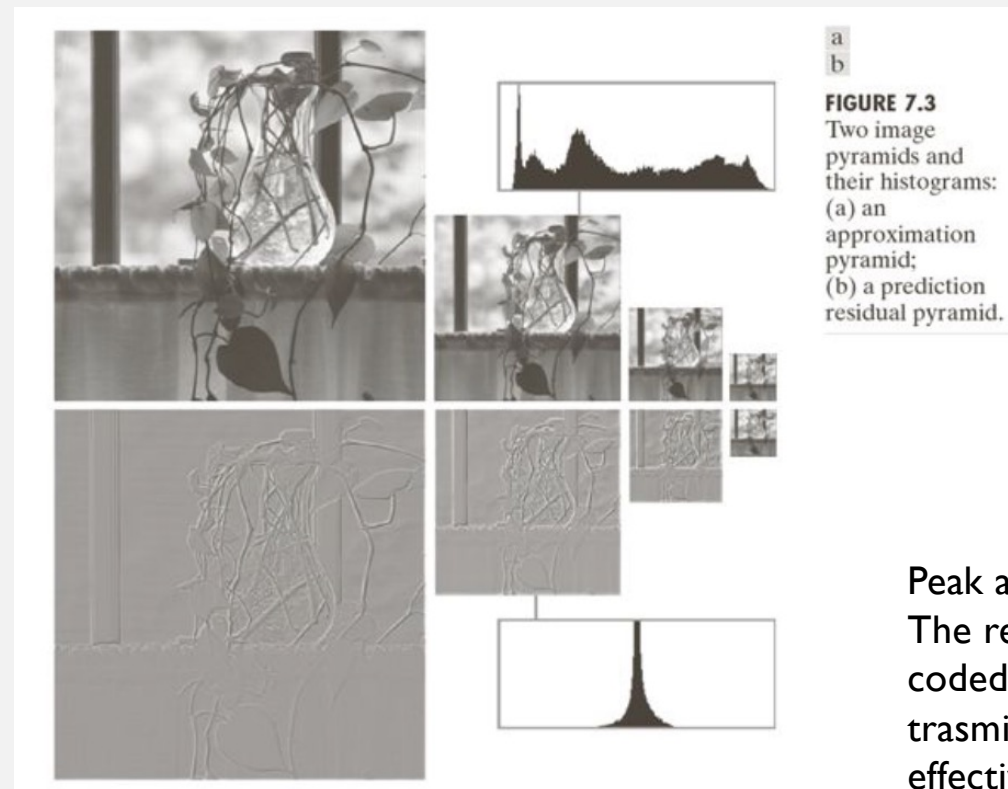


Image from Gonzalez & Woods, Digital Image Processing

Peak at around zero.
The residuals can be
coded, stored, and
transmitted more
effectively than
approximation

Use 5x5 low pass Gaussian
convolution kernel (for filtering)
-Original image 512×512 ($J = 9$)
- $P+1 = 4$ so $j = 6, 7, 8, 9$

Laplacian level 6 (64×64) predicts
Gaussian pyramid level 7 (128×128
approximation) by upsampling and
filtering and add the Laplacian level 7.

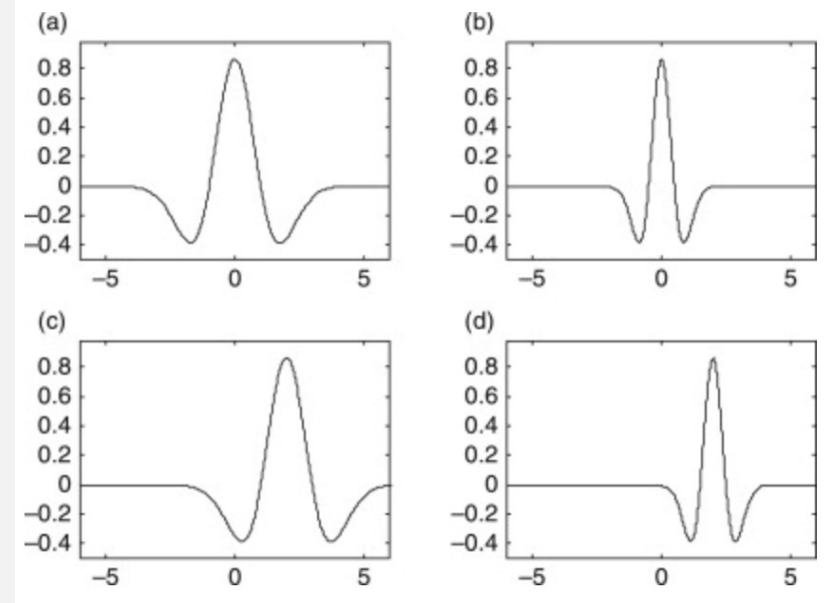
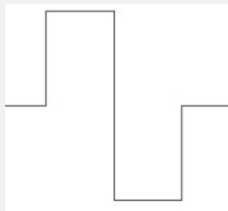
BASIS FUNCTIONS

- Basis functions used by the wavelet transform are all generated from one initial function called the mother wavelet. The simplest mother wavelet is the Haar wavelet. The basis functions are orthonormal wavelets.
- $h_\varphi(n)$ - scaling function coefficients
- $h_\psi(n)$ - wavelet function coefficients

$$h_\varphi(n) = \begin{cases} 1/\sqrt{2} & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$



$$h_\psi(n) = \begin{cases} 1/\sqrt{2} & n = 0 \\ -1/\sqrt{2} & n = 1 \\ 0 & \text{otherwise.} \end{cases}$$

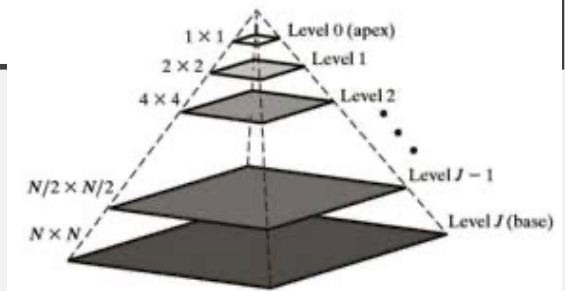


The Mexican Hat wavelet. (a) Mother wavelet, (b) dilation by a factor of 2, (c) translation of 4 to the right, and (d) combination of (b) and (c).

$$\psi(t) = \frac{2}{\sqrt{3}} \pi^{-\frac{1}{4}} (1 - t^2) e^{-\frac{t^2}{2}}$$

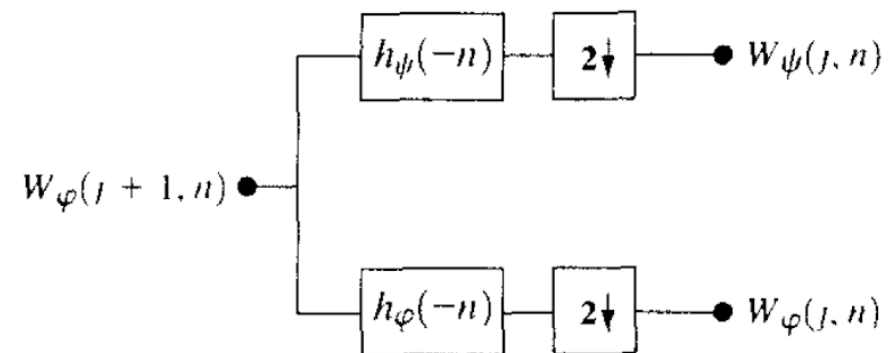
ONE-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)

- **FWT Analysis bank**
- Efficient implementation of discrete wavelet transform



$$W_{\psi}(j, k) = h_{\psi}(-n) * W_{\varphi}(j + 1, n) \Big|_{n=2k, k \geq 0}$$

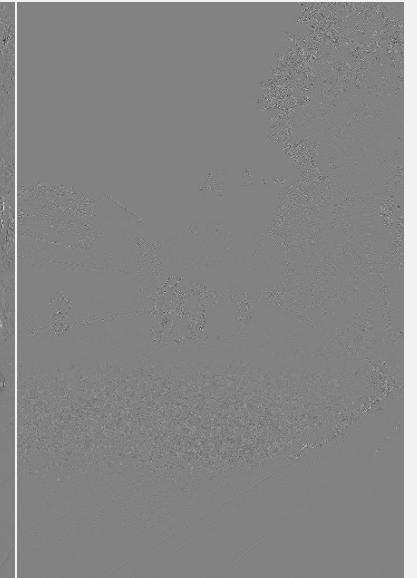
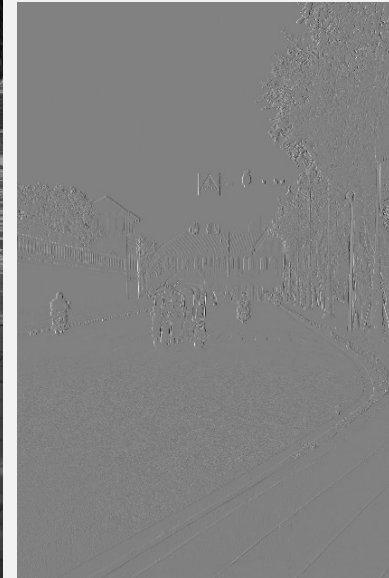
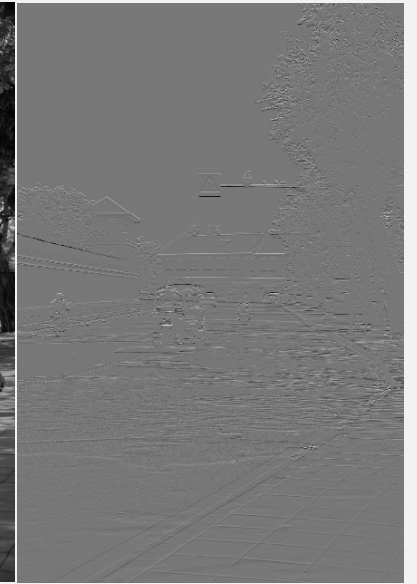
$$W_{\varphi}(j, k) = h_{\varphi}(-n) * W_{\varphi}(j + 1, n) \Big|_{n=2k, k \geq 0}$$



$W_{\varphi}(j, k)$ - approximation coefficients with scale level j , translating it by k . $W_{\psi}(j, k)$ - details coefficients. The image in level $j + 1$ is subsampled from level j so $2k = n$.

$$W_\psi(j, k) = h_\psi(-n) * W_\varphi(j + 1, n) \Big|_{n=2k, k \geq 0}$$

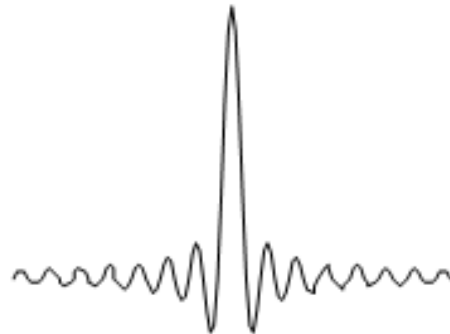
$$W_\varphi(j, k) = h_\varphi(-n) * W_\psi(j + 1, n) \Big|_{n=2k, k \geq 0}$$



OTHER WAVELETS



Haar



Shannon or Sinc



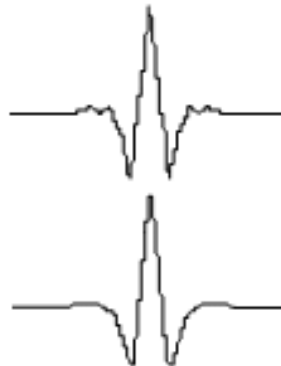
Daubechies 4



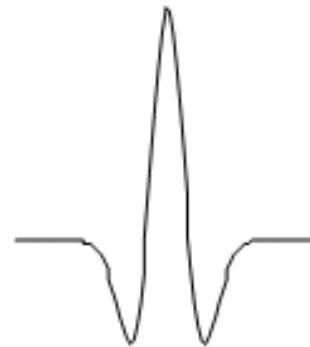
Daubechies 20



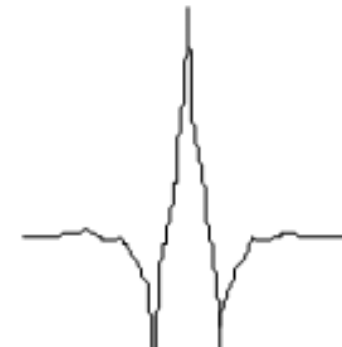
Gaussian or Spline



Biorthogonal



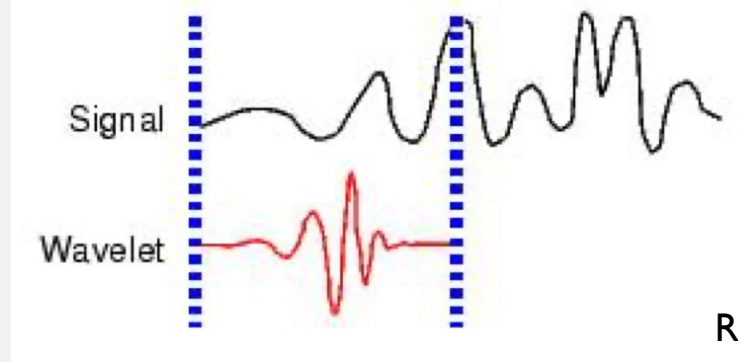
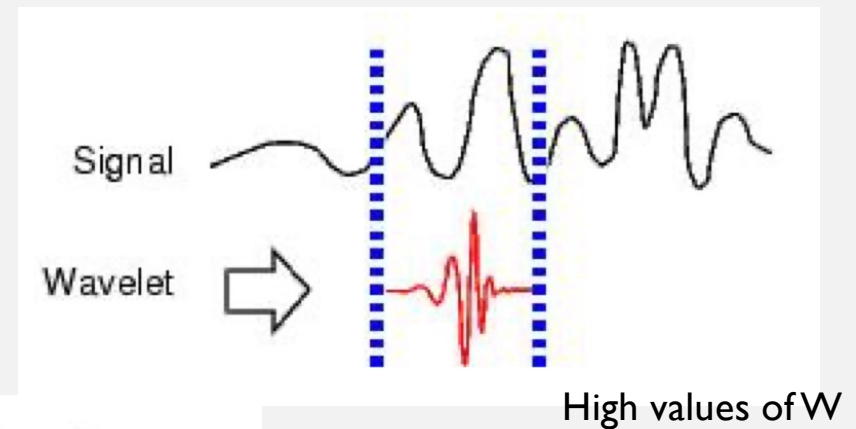
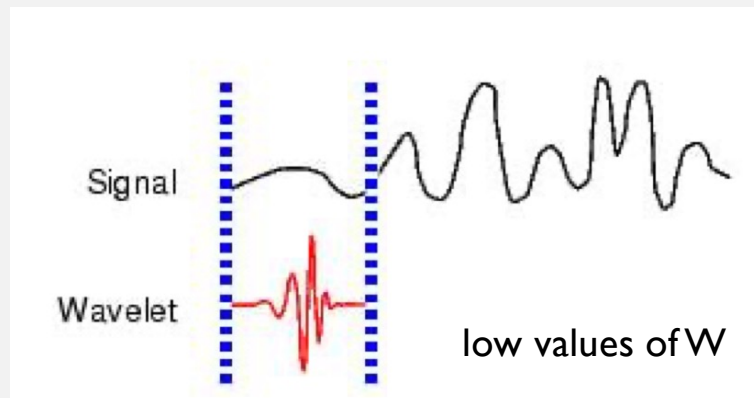
Mexican Hat



Coiflet

WAVELET TRANSFORM IN ONE DIMENSION

- **Continuous wavelet transform**



Reference: Colorado school of Mines

ONE-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)

- FWT Analysis bank**

- $f(n) = \{1, 4, -3, 0\}$ find DWT corresponding scaling and wavelet vectors (Haar)

$$h_{\varphi}(n) = \begin{cases} 1/\sqrt{2} & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

$$h_{\psi}(n) = \begin{cases} 1/\sqrt{2} & n = 0 \\ -1/\sqrt{2} & n = 1 \\ 0 & \text{otherwise.} \end{cases}$$

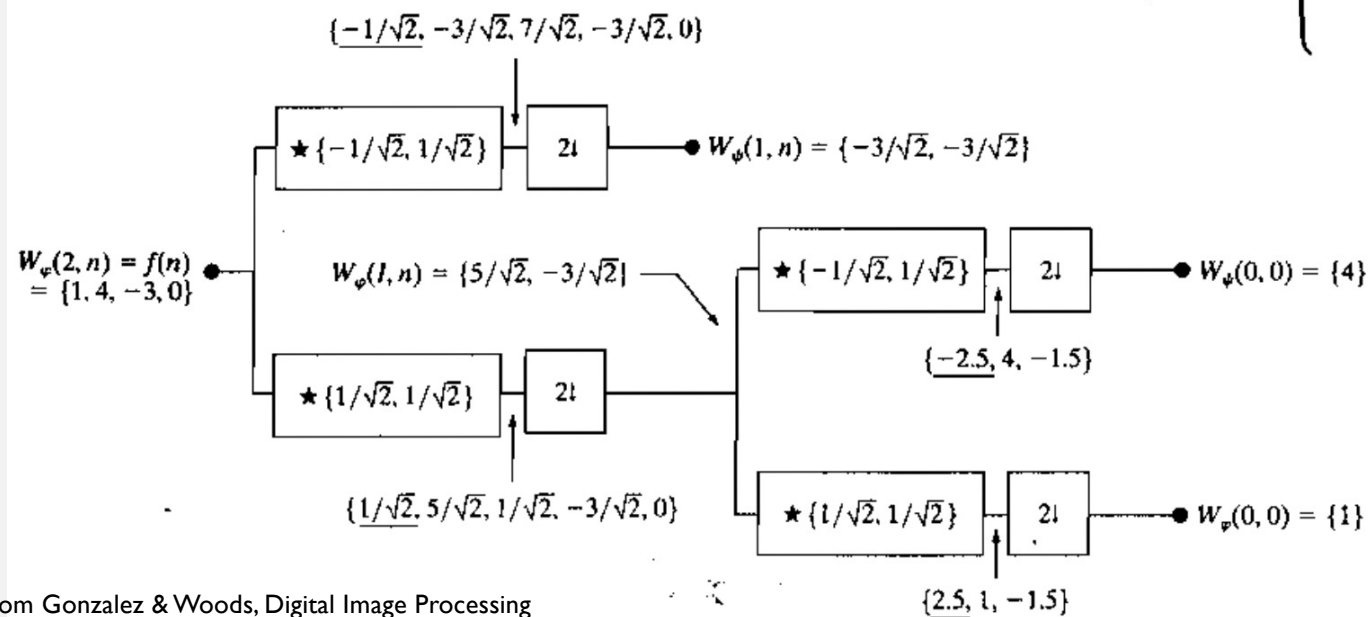
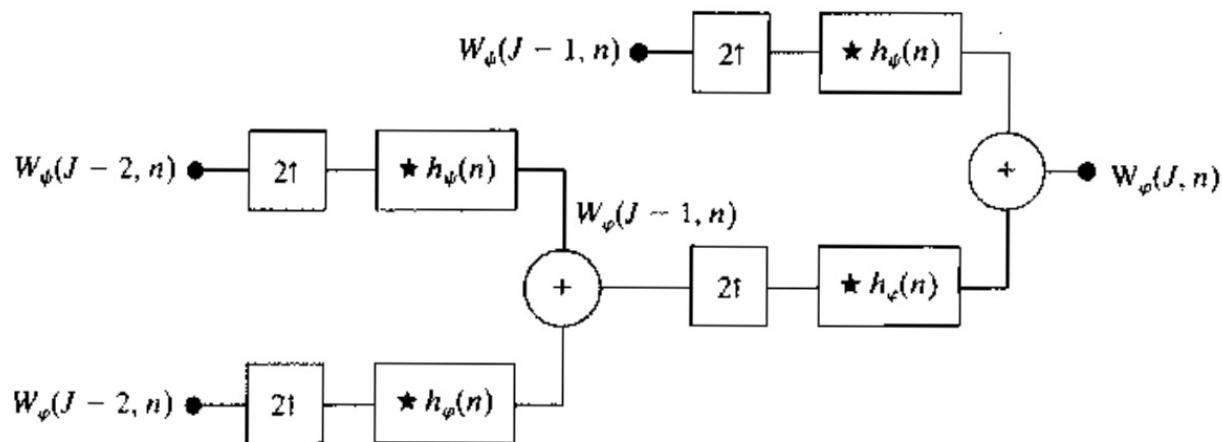


Image from Gonzalez & Woods, Digital Image Processing

ONE-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)



FWT Synthesis bank

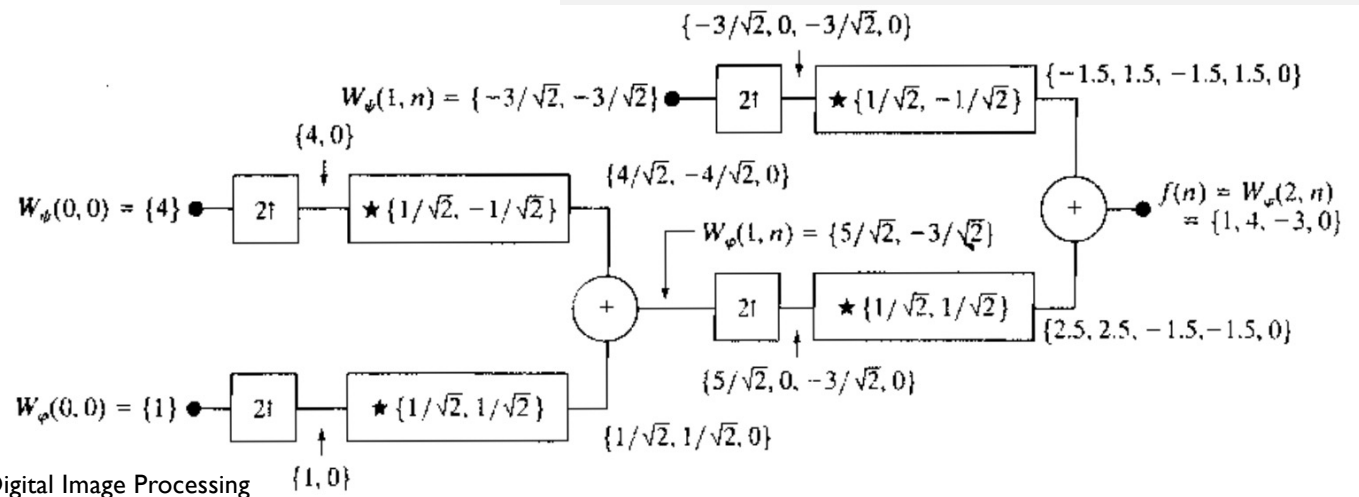


Image from Gonzalez & Woods, Digital Image Processing

TWO-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)

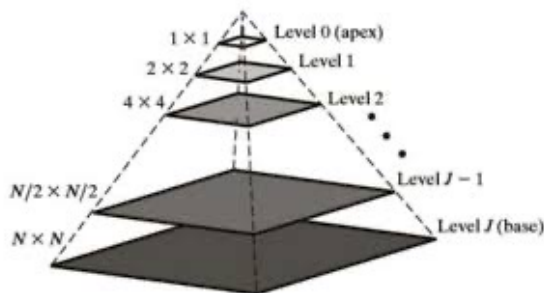
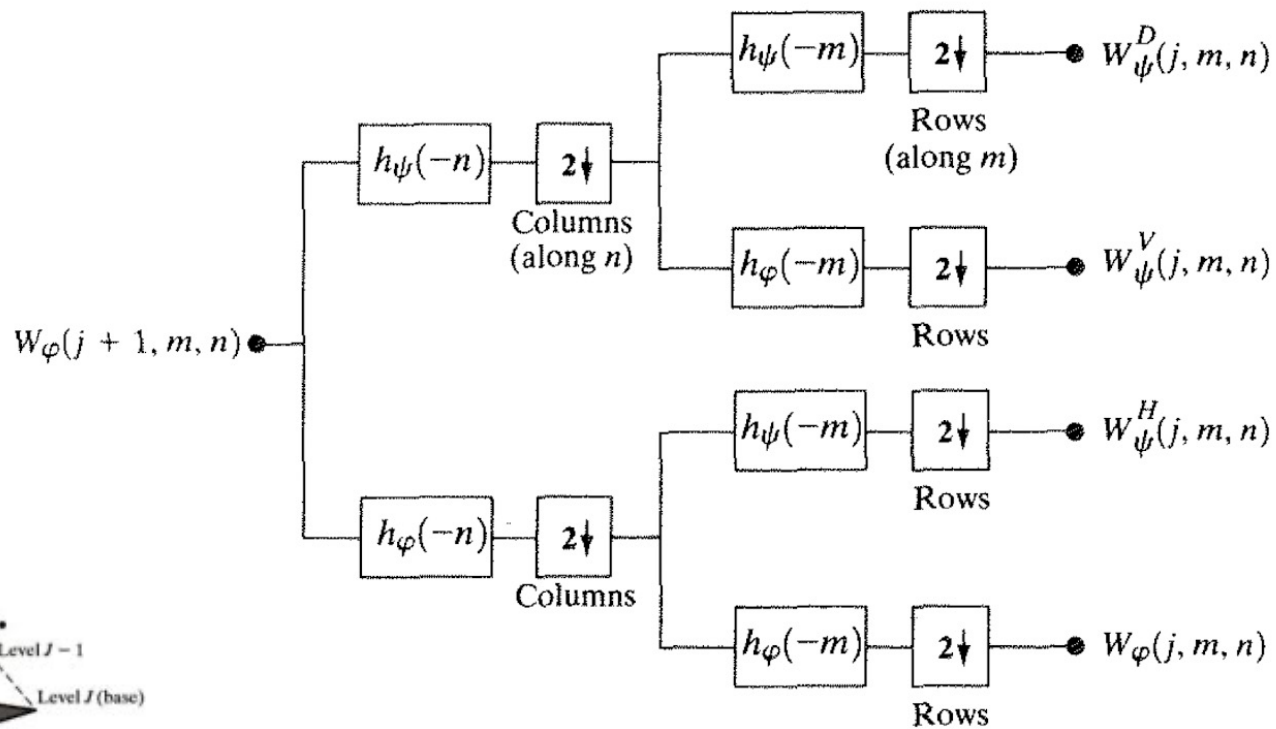


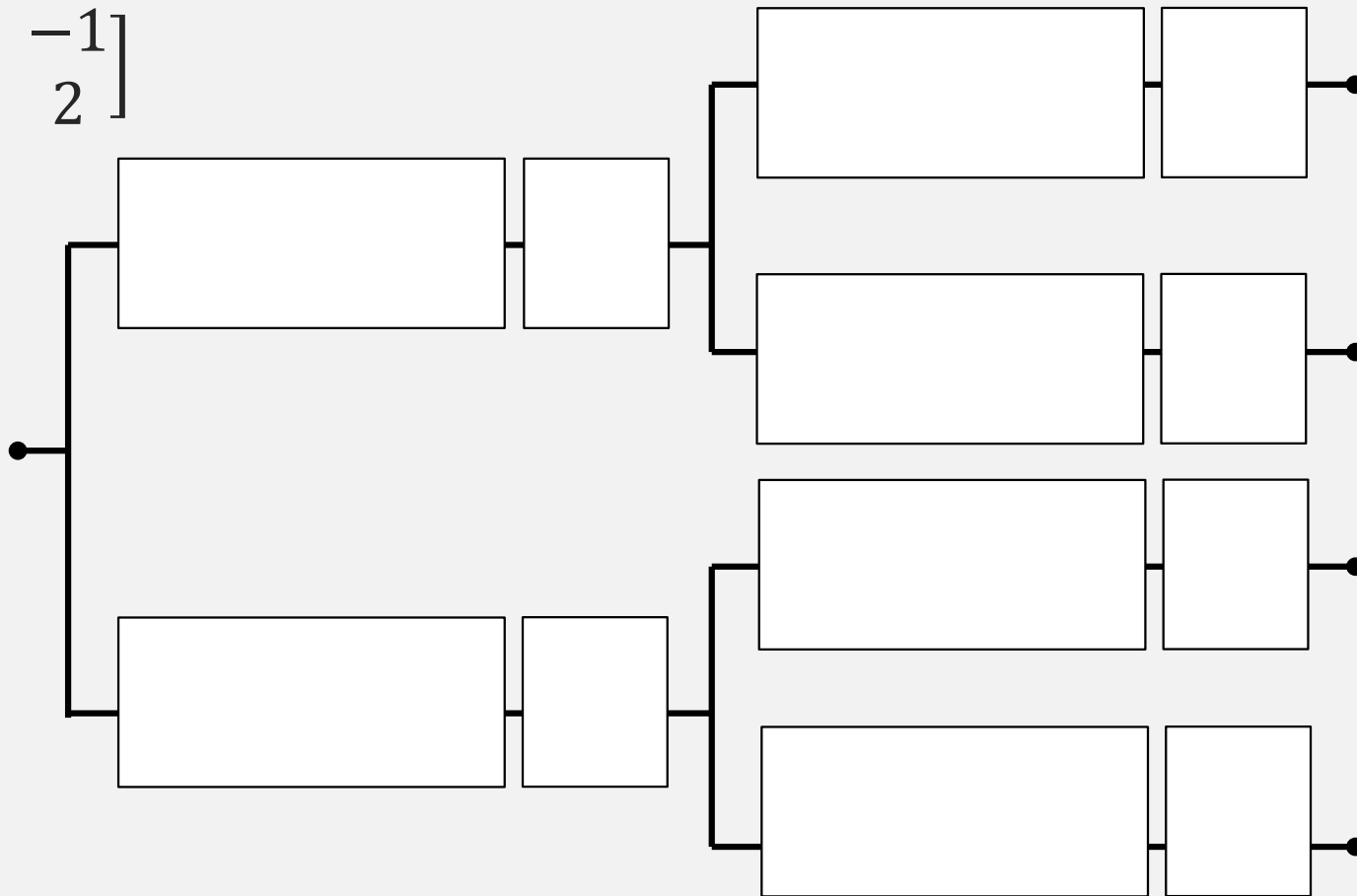
Image from Gonzalez & Woods, Digital Image Processing

TWO-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)

- Practice: a small image 2×2 $F = \begin{bmatrix} 3 & -1 \\ 6 & 2 \end{bmatrix}$
- Compute 2D Wavelet transform with respect to Haar wavelets of this image. Draw the required filter bank and label all inputs and outputs with proper arrays.

TWO-DIMENSIONAL FAST WAVELET TRANSFORM (FWT)

• $F = \begin{bmatrix} 3 & -1 \\ 6 & 2 \end{bmatrix}$



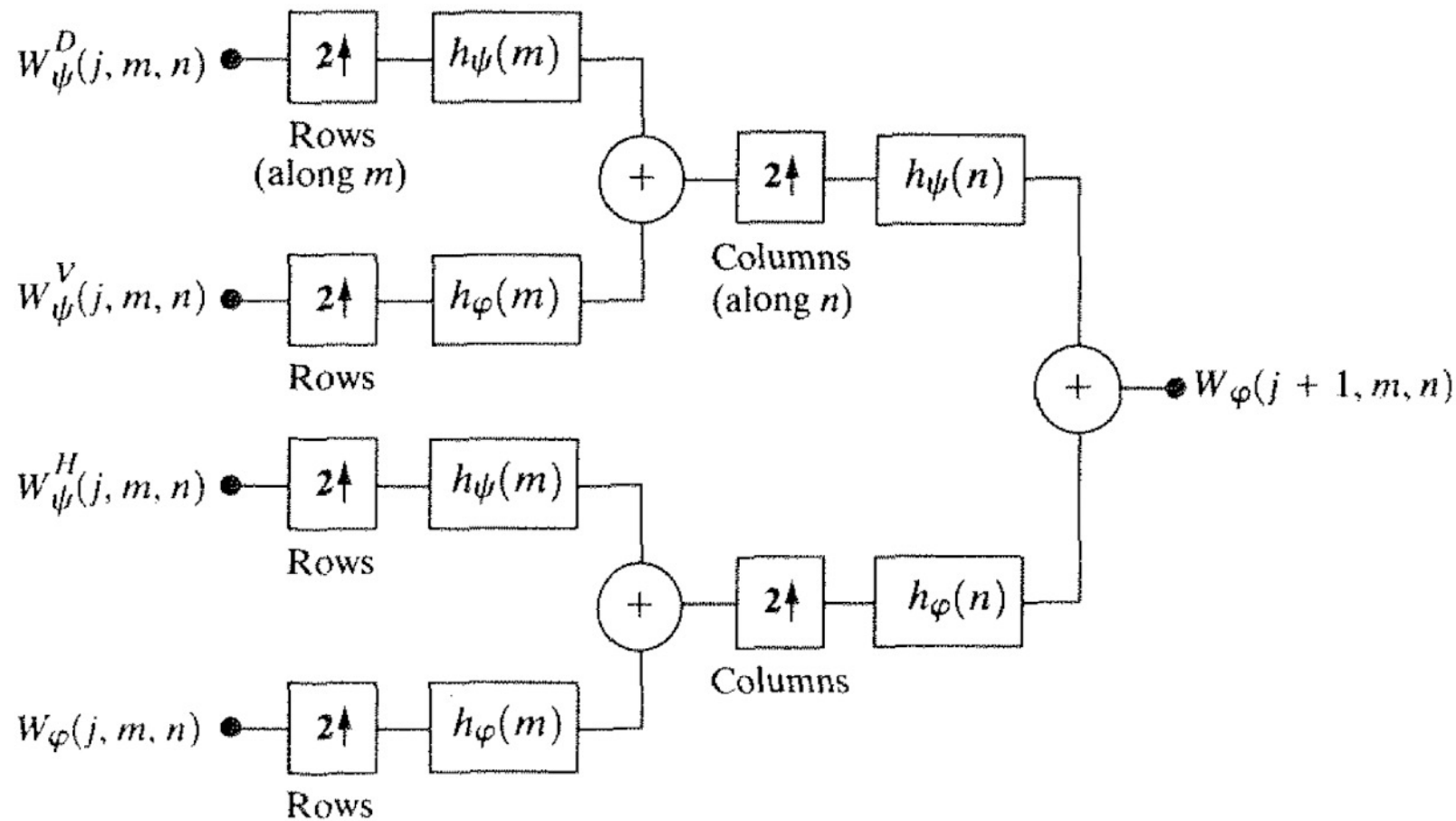
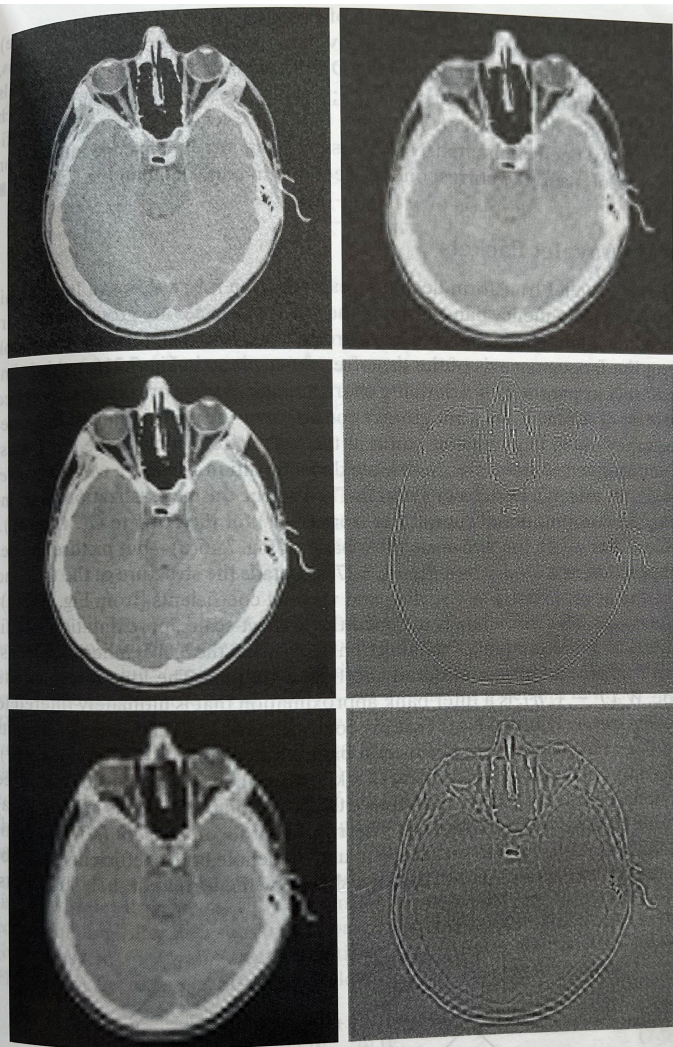


FIGURE 7.22 The two-dimensional fast wavelet transform: (a) the analysis filter bank; (b) the resulting decomposition; and (c) the synthesis filter bank.

Synthesis filter banks

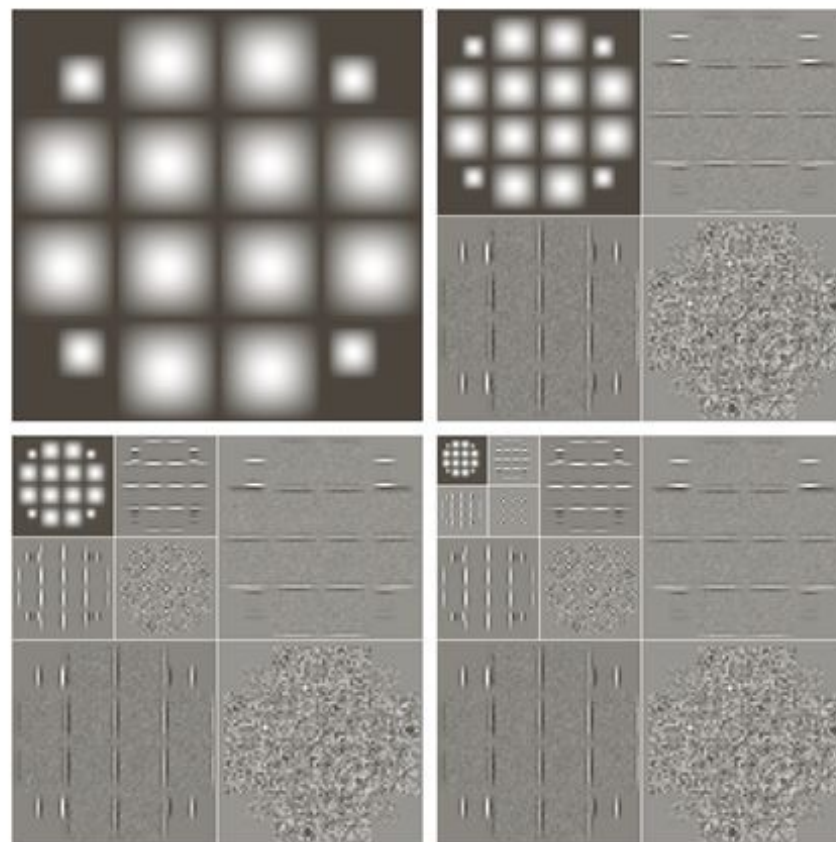
Image from Gonzalez & Woods, Digital Image Processing



a b
c d
e f

FIGURE 7.26
Modifying a DWT
for noise removal:
(a) a noisy MRI
of a human head;
(b), (c) and
(e) various
reconstructions
after thresholding
the detail
coefficients; (d)
and (f) the
information
removed during
the reconstruction
of (c) and (e).
(Original image
courtesy
Vanderbilt
University
Medical Center.)

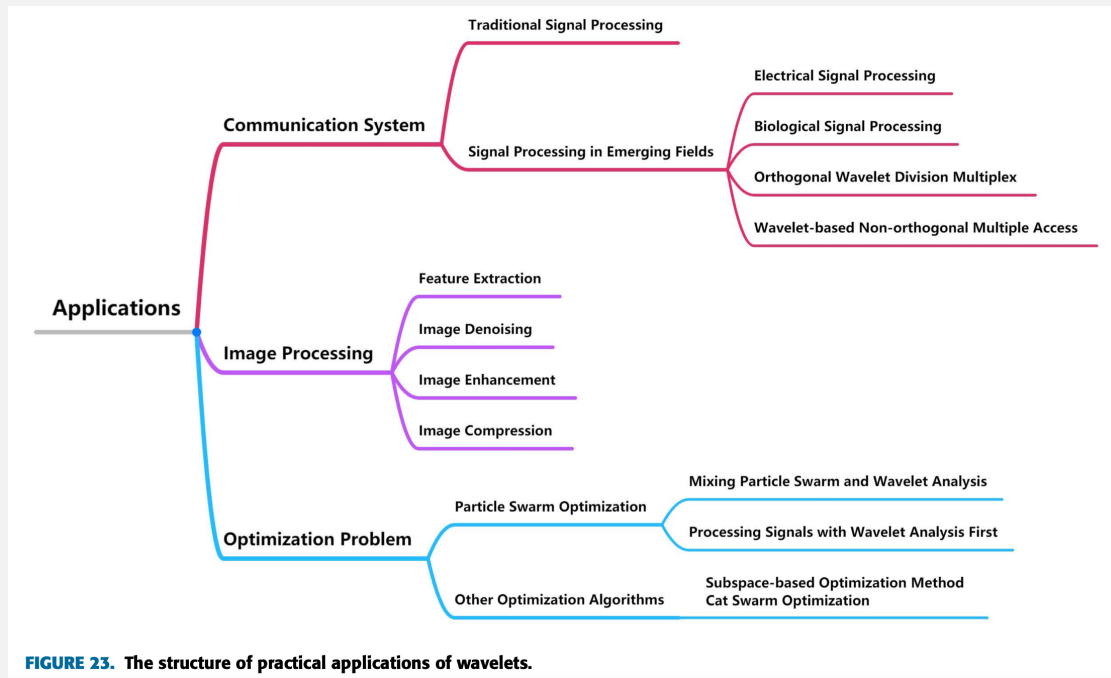
tails) and reconstructing the image. Here, almost all of the background noise has been eliminated and the edges are only slightly disturbed. Figure 7.26(d) shows the information that is lost in the process. This result was generated by computing the inverse FWT of the decomposed image with all but the highest-resolution detail coefficients zeroed. As can be seen, it contains most of the



a b
c d

FIGURE 7.25
Computing a 2-D
three-scale FWT:
(a) the original
image; (b) a one-
scale FWT; (c) a
two-scale FWT;
and (d) a three-
scale FWT.

APPLICATIONS OF WAVELETS



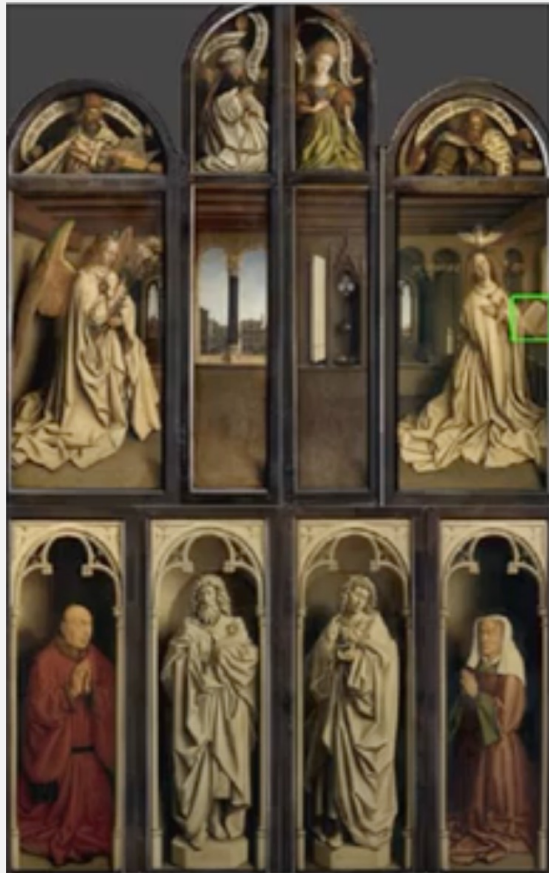
Wavelets for image processing

- Optimally extract features in different sizes
 - Palm- print images after two-level wavelet decomposition & maintaining the uniqueness of each palmprint image and can be used for palmprint image classification
- Denoising, enhancement and compression in the image processing area
- Extracting structure and noise information through wavelet transform and generating high-quality images through GAN
- Wavelet kernel-based neural network
 - Initialization
 - Used as the activation function

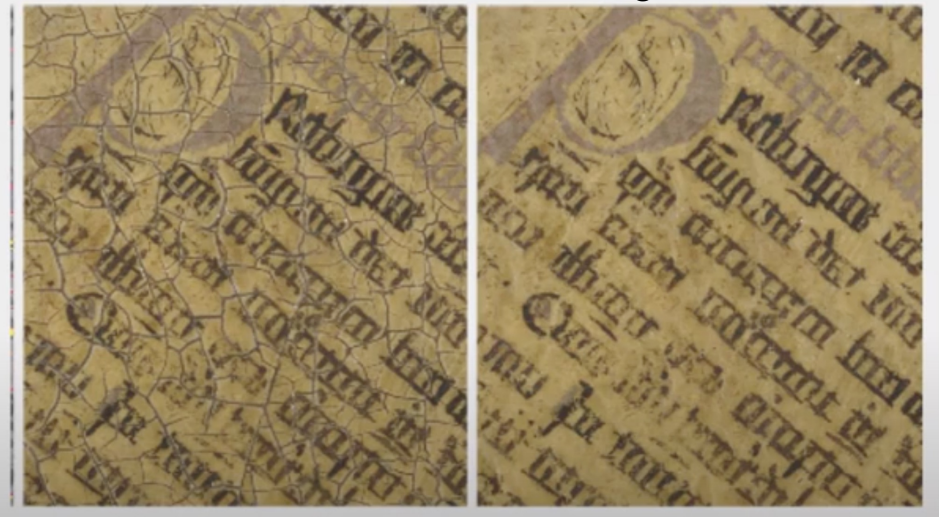
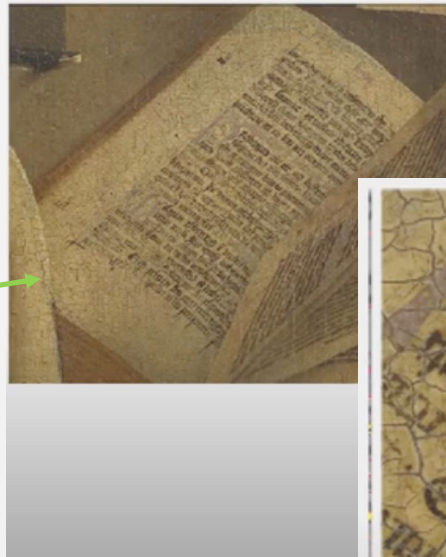
WAVELETS FOR ART



Ingrid Daubechies



Ghent Altarpiece



Ref. <https://www.youtube.com/watch?v=jrFISGPYF4g>
Using Math to Understand Art | Ingrid Daubechies | TEDxDuke

WAVELETS FOR AI

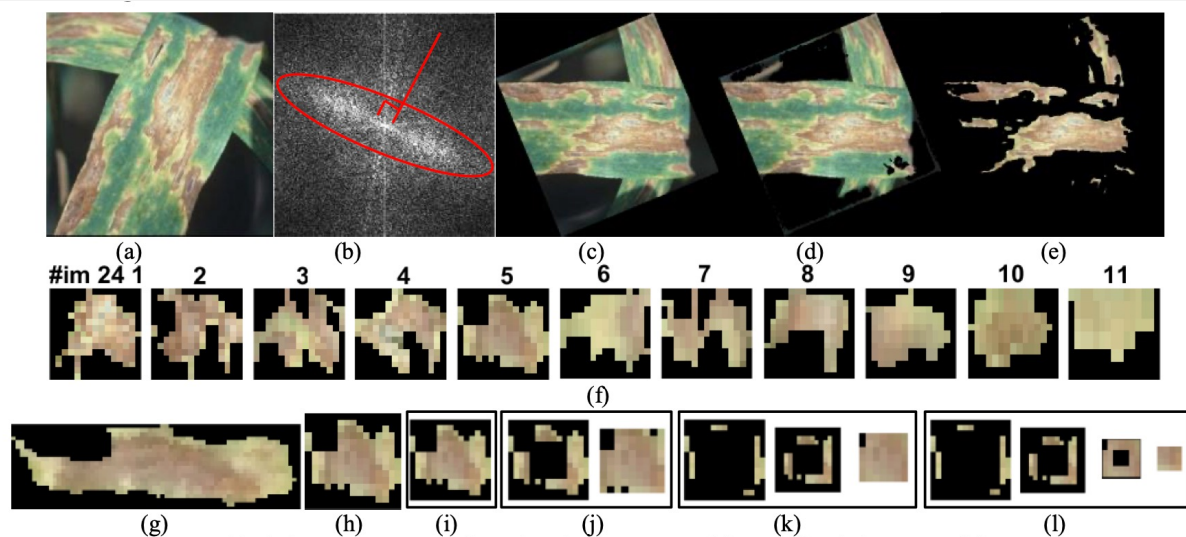


Fig. 4. Uncontrolled image processing in the system (a) original image, (b) DFT on Canny edge, (c) rotated image, (d) partially segmented image, (e) segmented disease, (f) constructed disease patches, (g) original disease blob (#5), (h) normalised disease patch (i) 1-level (j) 2-level (k) 3-level (l) 4-level radial pyramid patches

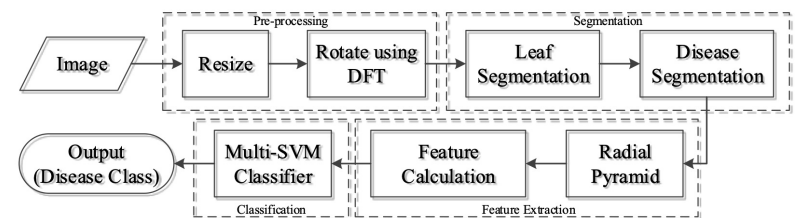


Fig. 1. Our proposed system for wheat disease classification

3.2 Leaf segmentation using multi-resolution discrete Wavelet transform

A single-level two-dimensional Wavelet transform (DWT) is exploited to decompose an image into coefficients of four different components, an approximation component (cA), horizontal, vertical, and diagonal details (cH , cV , and cD). We combined horizontal, vertical and diagonal detail information based on Daubechies wavelets and then thresholded the low coefficient values out to remove part of the background from consideration in the subsequent stages of our process (see Fig. 4 (d)).

Punnarai Siricharoen, Bryan W. Scotney, Philip J. Morrow, Gerard P. Parr:
Automated Wheat Disease Classification Under Controlled and Uncontrolled Image
Acquisition. ICIAR 2015: 456-464

EXERCISE #1 USE PYWAVELETS FOR IMAGE SPACE-FREQUENCY ANALYSIS

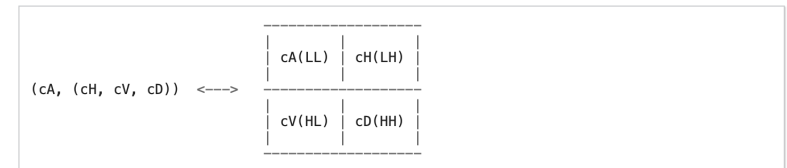
- Reference: <https://pywavelets.readthedocs.io/en/latest/>

```
import pywt
import cv2

original = cv2.imread("kitty.png", 0)
LL, (LH, HL, HH) = pywt.dwt2(original, 'haar')
```

```
# display
fig = plt.figure(figsize=(12, 3))
for i, a in enumerate([LL, LH, HL, HH]):
    ax = fig.add_subplot(1, 4, i + 1)
    ax.imshow(a, interpolation="nearest",
              cmap=plt.cm.gray)
    ax.set_title(titles[i], fontsize=10)
    ax.set_xticks([])
    ax.set_yticks([])
    fig.tight_layout()
plt.show()
```

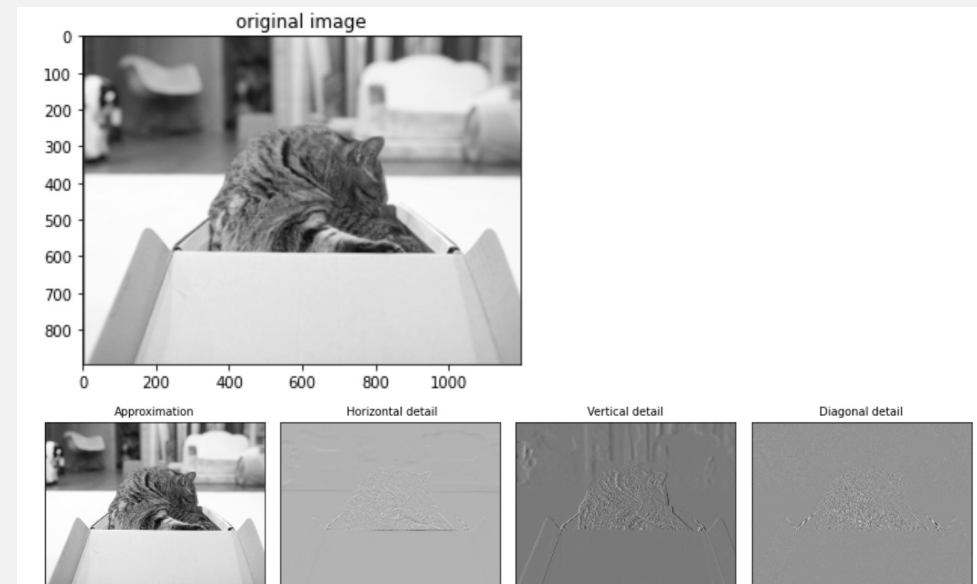
The relation to the other common data layout where all the approximation and details coefficients are stored in one big 2D array is as follows:



PyWavelets does not follow this pattern because of pure practical reasons of simple access to particular type of the output coefficients.

EXERCISE #1 USE PYWAVELETS FOR IMAGE SPACE-FREQUENCY ANALYSIS

- Investigate the size of the original image, approximation, horizontal, vertical and diagonal details.
- Investigate the min/max values in the approximation, horizontal, vertical and diagonal details.



Reference:

<https://pywavelets.readthedocs.io/en/latest/>

EXERCISE #2 CAN YOU SEGMENT THE CAT?

- 1) apply wavelet transform into "kitty.jpg" image
- 2) Again, investigate the min/max values in the approximation, horizontal, vertical and diagonal details.
- 3) Can you use a threshold to separate the high details from the low and segment the cat out?